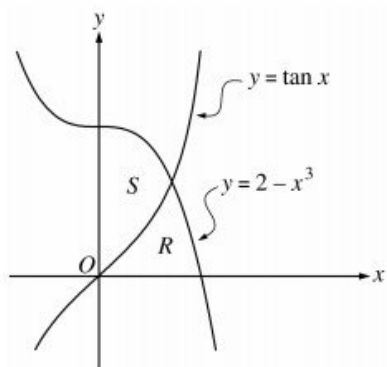


8-5-7

1.



Let R and S be the regions in the first quadrant shown in the figure above. The region R is bounded by the x -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$. The region S is bounded by the y -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$.



Find the area of R .

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

$$2 - x^3 = \tan x, \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area, } R = \int_0^A \tan x \, dx + \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx = 0.729$$

or

$$\text{Area, } R = \int_0^B ((2 - x)^{1/3} - \tan^{-1} y) \, dy = 0.729$$

or

$$\text{Area, } R = \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx - \int_0^A (2 - x^3 - \tan x) \, dx = 0.729$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

8-5-7

$$2 - x^3 = \tan x, \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area}, R = \int_0^A \tan x dx + \int_A^{\sqrt[3]{2}} (2 - x^3) dx = 0.729$$

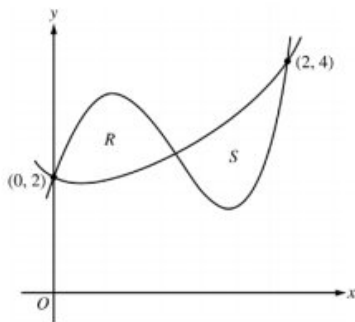
or

$$\text{Area}, R = \int_0^B \left((2 - x)^{1/3} - \tan^{-1} y \right) dy = 0.729$$

or

$$\text{Area}, R = \int_A^{\sqrt[3]{2}} (2 - x^3) dx - \int_0^A (2 - x^3 - \tan x) dx = 0.729$$

8-5-7



Let f and g be the functions defined by $f(x) = 1 + x + e^{x^2-2x}$ and $g(x) = x^4 - 6.5x^2 + 6x + 2$. Let R and S be the two regions enclosed by the graphs of f and g shown in the figure above.

2.  Find the sum of the areas of regions R and S .

Part A

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y=f(x)$ and $y=g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and $(A, B)=(1.032832, 2.401108)$

$$\text{Area} = \int_0^A [g(x) - f(x)] dx + \int_A^2 [f(x) - g(x)] dx$$

$$= 0.997427 + 1.006919 = 2.004$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y=f(x)$ and $y=g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and $(A, B)=(1.032832, 2.401108)$

$$\text{Area} = \int_0^A [g(x) - f(x)] dx + \int_A^2 [f(x) - g(x)] dx$$

$$= 0.997427 + 1.006919 = 2.004$$

8-5-7

3.  Region S is the base of a solid whose cross sections perpendicular to the x -axis are squares. Find the volume of the solid.

Part B

2 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$



0	1	2
---	---	---

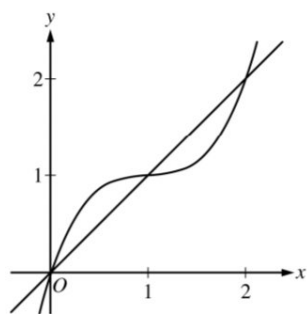
The student response earns all of the following points:

2 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$

4.



The graphs of the function g and the line $y = x$ are shown in the figure above. The graphs intersect at the points $(0, 0)$, $(1, 1)$, and $(2, 2)$.

Which of the following expressions give the area enclosed by the graphs?

I $\left| \int_0^2 (x - g(x)) dx \right|$

II $\int_0^2 |x - g(x)| dx$

III. $\int_0^1 (g(x) - x) dx + \int_1^2 (x - g(x)) dx$

(A) II only

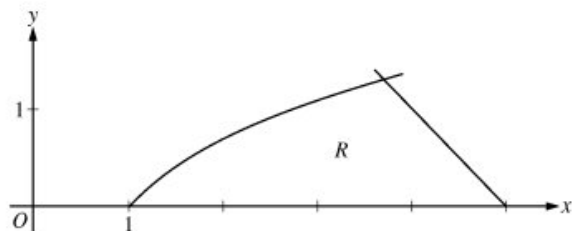
(B) III only

(C) I and II only

(D) II and III only



8-5-7



Let R be the region in the first quadrant bounded by the x -axis and the graphs of $y = \ln x$ and $y = 5 - x$, as shown in the figure above.

5. Find the area of R .

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\text{Area} = \int_0^B (5 - y - e^y) dy$$

$$= 2.986 \text{ (or } 2.985)$$

OR

$$\text{Area} = \int_1^A \ln x dx + \int_A^5 (5 - x)^2$$

$$= 2.986 \text{ (or } 2.985)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\text{Area} = \int_0^B (5 - y - e^y) dy$$

$$= 2.986 \text{ (or } 2.985)$$

8-5-7

OR

$$\text{Area} = \int_1^A \ln x dx + \int_A^5 (5-x)^2$$

$$= 2.986 \text{ (or } 2.985)$$

6. The horizontal line $y=k$ divides R into two regions of equal area. Write, but do not solve, an equation involving one or more integrals whose solution gives the value of k .

Part C

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the equation

$$\ln x = 5-x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y=\ln x$ and $y=5-x$ intersect in the first quadrant at the point $(A, B)=(3.69344, 1.30656)$.

$$\int_0^K (5-y-e^y) dy = \frac{1}{2} \cdot 2.986 \text{ (or } \frac{1}{2} \cdot 2.985)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the equation

$$\ln x = 5-x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y=\ln x$ and $y=5-x$ intersect in the first quadrant at the point $(A, B)=(3.69344, 1.30656)$

$$\int_0^K (5-y-e^y) dy = \frac{1}{2} \cdot 2.986 \text{ (or } \frac{1}{2} \cdot 2.985)$$

8-5-7

7. Region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part B

2 points are earned for the integrands

1 point is earned for the expression for total volume

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$

$$\text{Volume} = \int_1^A (\ln x)^2 dx + \int_A^5 (5 - x)^2 dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrands

1 point is earned for the expression for total volume

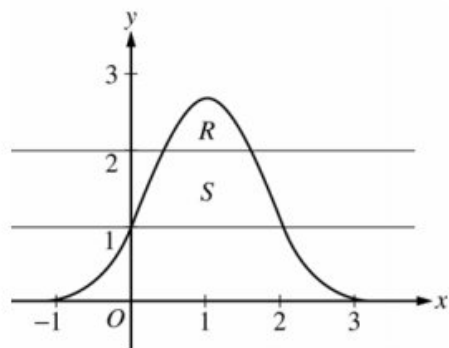
$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$

$$\text{Volume} = \int_1^A (\ln x)^2 dx + \int_A^5 (5 - x)^2 dx$$

8-5-7

8.



Let R be the region bounded by the graph of $y=e^{2x-x^2}$ and the horizontal line $y=2$, and let S be the region bounded by the graph of $y=e^{2x-x^2}$ and the horizontal lines $y=1$ and $y=2$, as shown above.



(b) Find the area of S .

Part B

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2}=2 \text{ when } x=0.446057, 1.553943$$

Let $P=0.446057$ and $Q=1.553943$

$$e^{2x-x^2}=1 \text{ when } x=0, 2$$

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1) dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

OR

$$\int_0^P (e^{2x-x^2} - 1) dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1) dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

8-5-7

1 point is earned for the answer

$$e^{2x-x^2}=2 \text{ when } x=0.446057, 1.553943$$

Let $P=0.446057$ and $Q=1.553943$

$$e^{2x-x^2}=1 \text{ when } x=0, 2$$

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1) dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

OR

$$\int_0^P (e^{2x-x^2} - 1) dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1) dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$

9. What is the area enclosed by the curves $y = x^3 - 8x^2 + 18x - 5$ and $y = x + 5$?

(A) 10.667

(B) 11.833

(C) 14.583

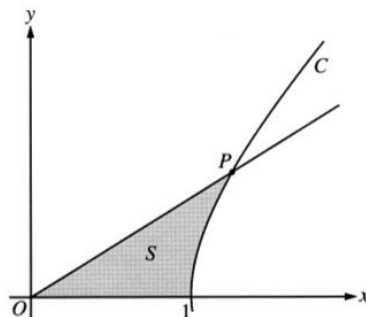
(D) 21.333

(E) 32



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10.



The figure above shows the graphs of the line $x = \frac{5}{3}y$ and the curve C given by $x = \sqrt{1 + y^2}$. Let S be the shaded region bounded by the two graphs and the x -axis. The line and the curve intersect at point P .



Set up and evaluate an integral expression with respect to y that gives the area of S .

Part B

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{Area} = \int_0^{3/4} (\sqrt{1 + y^2} - \frac{5}{3}y) dy$$

1 point is earned for the correct answer

$$= 0.346 \text{ or } 0.347$$



0	1	2	3
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The student response earns three of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

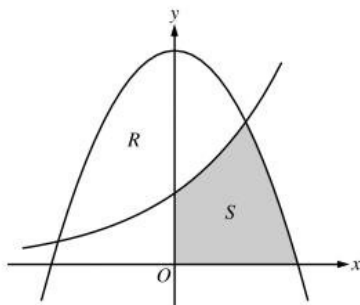
$$\text{Area} = \int_0^{3/4} (\sqrt{1 + y^2} - \frac{5}{3}y) dy$$

1 point is earned for the correct answer

$$= 0.346 \text{ or } 0.347$$

8-5-7

11.



Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.



Find the area of S .

Part A

1 point is earned for integrands

1 point is earned for limits

1 point is earned for the answer

$3 - x^2 = 2^x$ when $x = -1.63658$ and $x = 1$

Let $a = -1.63658$

$$\begin{aligned}\text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrands

1 point is earned for limits

1 point is earned for the answer

$3 - x^2 = 2^x$ when $x = -1.63658$ and $x = 1$

Let $a = -1.63658$

$$\begin{aligned}\text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240\end{aligned}$$

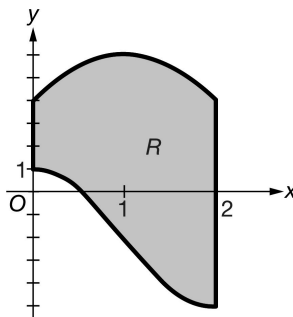
8-5-7

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let R be the region enclosed by the graphs of $g(x) = -2 + 3 \cos\left(\frac{\pi}{2}x\right)$ and $h(x) = 6 - 2(x-1)^2$, the y -axis, and the vertical line $x = 2$, as shown in the figure above.

- Find the area of R .
- Region R is the base of a solid. For the solid, at each x the cross section perpendicular to the x -axis has area $A(x) = \frac{1}{x+3}$. Find the volume of the solid.
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 6$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand that contains a form of $h(x) - g(x)$ or $g(x) - h(x)$.

The second point is earned for the antiderivative $\frac{6}{\pi} \sin\left(\frac{\pi}{2}x\right)$.

A response is eligible for the fourth point if (1) the first point is earned, (2) at least one of the second or third points is earned, (3) the integrand contains no copy errors or arithmetic errors. Simplification is not required to earn the fourth point. The limits of integration are part of the fourth point.

If the limits of integration are $x = 1$ to $x = 2$ in all parts of the question, the response is not eligible for the fourth point in Part A but eligible for points in Part B and Part C.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ integrand
- ☐ antiderivative of $3 \cos\left(\frac{\pi}{2}x\right)$
- ☐ antiderivative of remaining terms

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☐ answer
Solution:

$$\begin{aligned}\int_0^2 (h(x) - g(x)) \, dx &= \int_0^2 \left((6 - 2(x-1)^2) - \left(-2 + 3 \cos\left(\frac{\pi}{2}x\right)\right) \right) \, dx \\ &= \left[\left(6x - \frac{2}{3}(x-1)^3\right) - \left(-2x + \frac{6}{\pi} \sin\left(\frac{\pi}{2}x\right)\right) \right]_{x=0}^{x=2} \\ &= \left(\left(12 - \frac{2}{3}\right) - (-4 + 0) \right) - \left(\left(0 + \frac{2}{3}\right) - (0 + 0) \right) \\ &= 12 - \frac{2}{3} + 4 - \frac{2}{3} = \frac{44}{3}\end{aligned}$$

The area of R is $\frac{44}{3}$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for the definite integral $\int_0^2 A(x) \, dx$ or $\int_0^2 \frac{1}{x+3} \, dx$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\begin{aligned}\int_0^2 A(x) \, dx &= \int_0^2 \frac{1}{x+3} \, dx \\ &= [\ln(x+3)]_{x=0}^{x=2} = \ln 5 - \ln 3\end{aligned}$$

The volume of the solid is $\ln 5 - \ln 3$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for a definite integral, multiplied by π , with 0 and 2 as the lower and upper limits of integration, respectively. The π may appear in the integrand, but it must be correct to earn the first point.

The second point is earned with an integrand that is a difference of squares with a correct or consistent axis of rotation.

The third point is earned for the correct integrand. Note that a reversal of order within the squared terms, e.g.

$\pi \int_0^2 \left((g(x) - 6)^2 - (h(x) - 6)^2 \right) \, dx$, is equivalent to the given solution and thus earns the third point.



0	1	2	3
---	---	---	---

8-5-7

The student response accurately includes **all three** of the criteria below.

- ☐ limits and constant
- ☐ form of integrand
- ☐ integrand

Solution:

$$\pi \int_0^2 \left((6 - g(x))^2 - (6 - h(x))^2 \right) dx$$

8-5-7

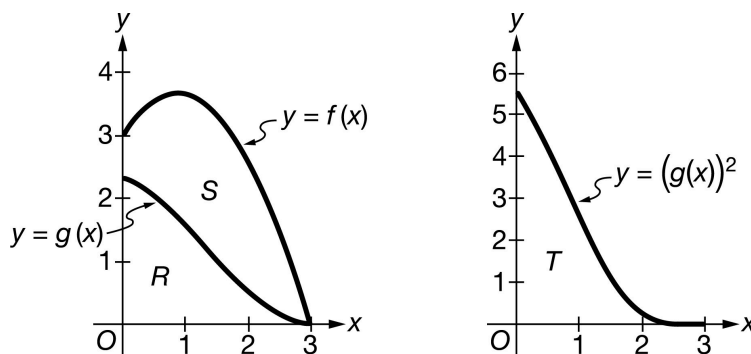
13. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The function f is defined by $f(x) = 3(1+x)^{0.5} \cos\left(\frac{\pi x}{6}\right)$ for $0 \leq x \leq 3$. The function g is continuous and decreasing for $0 \leq x \leq 3$ with $g(3) = 0$.

The figure above on the left shows the graphs of f and g and the regions R and S . R is the region bounded by the graph of g and the x - and y -axes. Region R has area 3.24125. S is the region bounded by the y -axis and the graphs of f and g .

The figure above on the right shows the graph of $y = (g(x))^2$ and the region T . T is the region bounded by the graph of $y = (g(x))^2$ and the x - and y -axes. Region T has area 5.32021.



(a) Find the area of region S .

(b) Find the volume of the solid generated when region S is revolved about the horizontal line $y = -3$.

(c) Region S is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 7 times the length of its base in region S . Write, but do not evaluate, an integral expression for the volume of this solid.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned with either $\int_0^3 (f(x) - g(x)) \, dx$ or $\int_0^3 f(x) \, dx - 3.24125$.

The second point is earned for evidence of 3.24125 being used as an area, which means it is being subtracted outside of an integrand. For example, $\int_0^3 f(x) \, dx - 3.24125$ would earn the point, while $\int_0^3 f(x) - 3.24125 \, dx$ would provide evidence for earning this point only in the presence of the final answer of 4.920 (or 4.919).

Note that responses in degree mode, which lead to a final answer of approximately 10.756727, do not earn the first point for which they are eligible. Thus a fully correct solution in degree mode would earn two of the three points.

8-5-7

✓

0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ definite integral of f
- ☐ uses area of R
- ☐ answer

Solution:

$$\begin{aligned}\int_0^3 (f(x) - g(x)) \, dx &= \int_0^3 f(x) \, dx - \int_0^3 g(x) \, dx \\ &= \int_0^3 f(x) \, dx - 3.24125 = 4.919585\end{aligned}$$

The area of region S is 4.920 (or 4.919).

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for an integrand of the form $(f(x) - k)^2 - (g(x) - k)^2$ where $k = -3$, 0 , or 3 .

The second point can be earned if f and g are reversed.

The third point is earned for replacing $\int_0^3 g(x) \, dx$ with 3.24135 and $\int_0^3 (g(x))^2 \, dx$ with 5.32021. There are multiple ways that a response can use these values correctly, so the third point can also be earned with sufficient work in the presence of the correct final value of 156.263709.

✓

0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ form of integrand
- ☐ integrand
- ☐ uses areas of R and T
- ☐ limits, constant, and answer

Solution:

8-5-7

$$\begin{aligned}
 & \pi \int_0^3 \left((f(x) + 3)^2 - (g(x) + 3)^2 \right) dx \\
 &= \pi \left(\int_0^3 (f(x) + 3)^2 dx - \int_0^3 \left((g(x))^2 + 6g(x) + 9 \right) dx \right) \\
 &= \pi \left(\int_0^3 (f(x) + 3)^2 dx - \int_0^3 (g(x))^2 dx - 6 \int_0^3 g(x) dx - \int_0^3 9 dx \right) \\
 &= \pi \left(\int_0^3 (f(x) + 3)^2 dx - 5.32021 - 6 \cdot 3.24125 - 9 \cdot 3 \right) \\
 &= 156.263709
 \end{aligned}$$

The volume of the solid is 156.264 (or 156.263).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an integrand of the form $(f(x) - g(x))^2$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integrand
- ☐ expression

Solution:

$$\text{Volume} = \int_0^3 7(f(x) - g(x))^2 dx$$

8-5-7

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

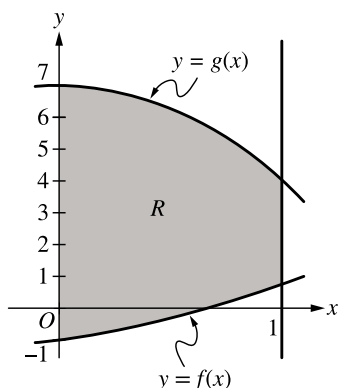


Figure 1

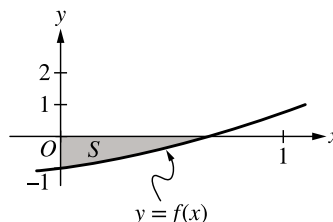


Figure 2

Let f and g be the functions given by $f(x) = e^x - 2$ and $g(x) = -3x^2 + 7$. Let R be the shaded region bounded by the graphs of f and g , the y -axis, and the vertical line $x = 1$, as shown in Figure 1. Let S be the shaded region in Quadrant IV bounded by the graph of f , the x -axis, and the y -axis, as shown in Figure 2.

- Find the area of R .
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 8$.
- The region S is the base of a solid. For this solid, each cross section perpendicular to the y -axis is a rectangle with height 5 and base in region S . Write, but do not evaluate, an integral expression that gives the volume of this solid.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand of $g(x) - f(x)$, $f(x) - g(x)$, $|g(x) - f(x)|$, or $|f(x) - g(x)|$ regardless of any subsequent copy errors in presentation.

- A response that presents any subsequent copy or parentheses errors is eligible for the second and third points, but not the fourth.

A response earns the fourth point only if it has earned the first 3 points and presents an answer equivalent to $9 - e$.

A response that uses an integrand of $f(x) - g(x)$:

- can earn the fourth point if it presents correct communication, such as $\int_0^1 (f(x) - g(x)) \, dx = e - 9$, so the area is $9 - e$.
- does not earn the fourth point if it includes incorrect communication, such as $\int_0^1 (f(x) - g(x)) \, dx = 9 - e$.

Special Case: A response that does not earn the first point, but presents $\int_0^1 (-3x^2 + 7 - e^x - 2) \, dx$ (or equivalent), is eligible for the second and third points, and earns the fourth point only for an answer of $5 - e$.

8-5-7

✓

0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ Integrand
- ☐ Antiderivative of $3x^2$ or e^x
- ☐ Remaining antiderivatives
- ☐ Answer

Solution:

$$\begin{aligned}
 \text{Area} &= \int_0^1 (g(x) - f(x)) \, dx \\
 &= \int_0^1 ((-3x^2 + 7) - (e^x - 2)) \, dx = \int_0^1 (-3x^2 + 9 - e^x) \, dx \\
 &= [-x^3 + 9x - e^x]_0^1 = (-1 + 9 - e) - (-1) = 9 - e
 \end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting both the correct constant and correct limits of integration in an integral expression involving both $f(x)$ and $g(x)$.

The second point is earned by an integrand of the form $R^2 - r^2$, where one of R and r is of the form $(\pm a \pm f(x))$ and the other is of the form $(\pm a \pm g(x))$, for $a \neq 0$.

The third point is earned only for the correct integrand.

✓

0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits and constant
- ☐ Form of integrand
- ☐ Integrand

Solution:

$$\text{Volume} = \pi \int_0^1 ((8 - f(x))^2 - (8 - g(x))^2) \, dx$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Alternate solution:

8-5-7

$$f(x) = 0 \Rightarrow x = \ln 2$$

$$\text{Volume} = 5 \cdot \text{Area}(S) = 5 \int_0^{\ln 2} (-f(x)) \, dx$$

These responses earn 1 of the 2 points:

- $5 \int_{-1}^0 \ln(y+2) \, dx$
- $5 \int_0^{\ln 2} (-f(x)) \, dy,$
- $5 \int_0^{\ln 2} -(e^x - 2) \, dy$
- $5 \int_{-1}^0 \ln(y+2)$
- $5 \int_0^{\ln 2} -f(x), 5 \int_0^{\ln 2} -(e^x - 2)$
- $5 \int_{-1}^0 -f(x) \, dy$
- $5 \int_0^{\ln 2} \ln(y+2) \, dx$

✓

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Limits and constant
- ☐ Integrand

Solution:

$$y = f(x) = e^x - 2 \Rightarrow x = \ln(y+2)$$

$$\text{Volume} = \int_{-1}^0 5 \ln(y+2) \, dy$$

8-5-7

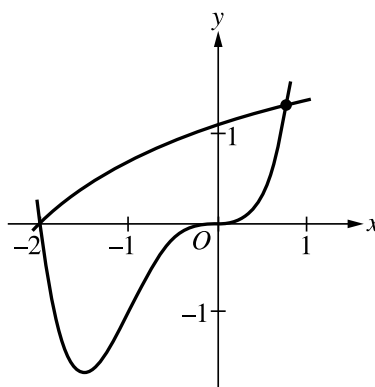
15.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f and g be the functions defined by $f(x) = \ln(x+3)$ and $g(x) = x^4 + 2x^3$. The graphs of f and g , shown in the figure, intersect at $x = -2$ and $x = B$, where $B > 0$.

- Find the area of the region enclosed by the graphs of f and g .
- For $-2 \leq x \leq B$, let $h(x)$ be the vertical distance between the graphs of f and g . Is h increasing or decreasing at $x = -0.5$? Give a reason for your answer.
- The region enclosed by the graphs of f and g is the base of a solid. Cross sections of the solid taken perpendicular to the x -axis are squares. Find the volume of the solid.
- A vertical line in the xy -plane travels from left to right along the base of the solid described in part (c). The vertical line is moving at a constant rate of 7 units per second. Find the rate of change of the area of the cross section above the vertical line with respect to time when the vertical line is at position $x = -0.5$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Other forms of the integrand in a definite integral, e.g., $|f(x) - g(x)|$, $|g(x) - f(x)|$, or $g(x) - f(x)$, earn the first point.

To earn the second point, the response must have a lower limit of -2 and an upper limit expressed as either the letter B with no value attached, or a number that is correct to the number of digits presented, with at least one and up to three decimal places.

• Case 1: If the response did not earn the second point because of an incorrect value of B , $0 < B < 1$, but used a lower limit of -2 , the response earns the third point only for a consistent answer.

• Case 2: If the response did not earn the second point because the lower limit used was $x = 0$, but the response used a correct upper limit of B the response earns the third point for a consistent answer of 0.708 (or 0.707).

8-5-7

· Case 3: If a response uses any other incorrect limits it does not earn the second or third points.

A response containing the integrand $g(x) - f(x)$ must interpret the value of the resulting integral correctly to earn the third point. For example, the following response earns all 3 points: $\int_{-2}^B (g(x) - f(x)) dx = -3.604$ so the area is 3.604. However, the response “

Area = $\int_{-2}^B (g(x) - f(x)) dx = 3.604$ ” presents an untrue statement and earns the first and second points but not the third point.

A response must earn the first point in order to be eligible for the third point. If the response has earned the second point, then only the correct answer will earn the third point.

Instructions for scoring a response that presents an integrand of $\ln(x+3) - x^4 + 2x^3$:

· The first time a response implicitly presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ (with no explicit connection) in any part of this question, the response loses a point. The response is then eligible for all remaining points for a correct or consistent answer.

· In part (a) the consistent answer using $j(x)$ is negative and will not earn the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integrand
- ☐ Limits of integration
- ☐ Answer

Solution:

$$\ln(x+3) = x^4 + 2x^3 \Rightarrow x = -2, x = B = 0.781975$$

$$\int_{-2}^B (f(x) - g(x)) dx = 3.603548$$

The area of the region is 3.604 (or 3.603).

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response need not present the value of $h'(-0.5)$. The last line earns both points. However, if a value is presented it must be correct for the digits reported up to three decimal places.

A response that reports an incorrect value of $h'(-0.5)$ earns only the first point.

A response that presents only $h'(x)$ does not earn either point.

A response that compares the values of $f'(x)$ and $g'(x)$ at $x = -0.5$ earns the first point and is eligible for the second point. This comparison can be made symbolically or verbally; for example, the response “the rate of change of $f(x)$ is less than the rate of change of $g(x)$ at $x = -0.5$ ” earns the first point.

Instructions for scoring a response that presents $h(x)$ as $\ln(x+3) - x^4 + 2x^3$:

· If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (b), the response loses a point. If the response already lost a point for this error in part (a), then the response is eligible for all remaining points for a correct or consistent answer.

8-5-7

· In part (b), the consistent answer using $j(x)$ is that $j'(-0.5) = 2.4 > 0$, so h is increasing at $x = -0.5$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(-0.5)$ – OR – $f'(x) - g'(x)$
- ☐ Answer with reason

Solution:

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(-0.5) = f'(-0.5) - g'(-0.5) = -0.6 \text{ (or } -0.599)$$

Since $h'(-0.5) < 0$, h is decreasing at $x = -0.5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand of $k(f(x) - g(x))^2$ or its equivalent with $k \neq 0$ in any definite integral. If $k \neq 1$, then the response is not eligible for the second point.

A response that does not earn the first point is ineligible to earn the second point, with the following exceptions:

· A response which has a presentation error in the integrand (for example, mismatched or missing parentheses, misplaced exponent) does not earn the first point but would earn the second point for the correct answer. A response which has a presentation error in the integrand and which reports an incorrect answer earns no points.

· A response that presents an integrand of $(\ln(x+3) - x^4 + 2x^3)^2$.

If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x+3) - x^4 + 2x^3$ occurs in part (c), the response loses a point. If the response already lost a point for this error in part (a) or part (b), then the response is eligible for all remaining points for a correct or consistent answer.

In part (c), the consistent answer using $j(x)$ is 252.187 252.187 (or 252.188).

A response that uses incorrect limits is only eligible for the second point, provided the limits are imported from part (a) in Case 1 or Case 2. In both of these situations, the second point is earned only for answers consistent with the imported limits.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Integrand
- ☐ Answer

Solution:

8-5-7

$$\int_{-2}^B (f(x) - g(x))^2 dx = 5.340102$$

The volume of the solid is 5.340.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point may be earned by presenting $\frac{dA}{dx} \cdot \frac{dx}{dt}$, $A' \cdot \frac{dx}{dt}$, $A'(x) \cdot \frac{dx}{dt}$, $A'(x) \cdot x'$, $A'(-0.5) \cdot 7$, or $k \cdot 7$, where k is a declared value of $A'(-0.5)$, or any equivalent expression, including $2(f(x) - g(x))(f'(x) - g'(x))\frac{dx}{dt}$.

If a response defines $f(x) - g(x)$ as a function in parts (b) or (c) (for example, $h(x) = f(x) - g(x)$), then a correct expression for $\frac{dA}{dt}$ (for example, $2h\frac{dh}{dt}$) earns the first point.

A response that imports a function $A(x)$ declared in part (c) is eligible for both points (the answer must be consistent with the imported function $A(x)$).

A response that presents an incorrect function for $A(x)$ that is not imported from part (c) is eligible only for the first point.

Except when $A(x)$ is imported from part (c), the second point is earned only for the correct answer.

A response that does not earn the first point is ineligible to earn the second point except in the special case noted below.

If the first time the response presents $f(x) - g(x)$ as $j(x) = \ln(x + 3) - x^4 + 2x^3$ occurs in part (d), the response loses a point. If the response already lost a point for this error in part (a), part (b), or part (c), then the response is eligible for all remaining points for a correct or consistent answer. In part (d) the consistent answer using $j(x)$ is 20.287.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\frac{dA}{dx} \cdot \frac{dx}{dt}$
- ☐ Answer

Solution:

The cross section has area $A(x) = (f(x) - g(x))^2$.

$$\frac{d}{dt}[A(x)] = \frac{dA}{dx} \cdot \frac{dx}{dt}$$

$$\left. \frac{d}{dt}[A(x)] \right|_{x=-0.5} = A'(-0.5) \cdot 7 = -9.271842$$

At $x = -0.5$, the area of the cross section above the line is changing at a rate of -9.272 (or -9.271) square units per second.

8-5-7

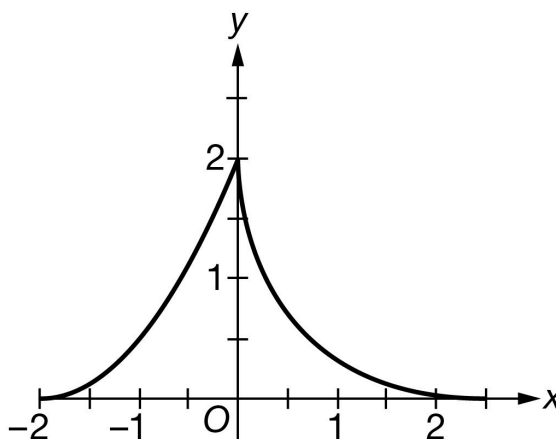
16.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

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Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be the function defined by $f(x) = \begin{cases} \frac{1}{2}(x+2)^2 & \text{for } -2 \leq x < 0 \\ 2 - 2\sin\sqrt{x} & \text{for } 0 \leq x \leq \frac{\pi^2}{4} \end{cases}$. The graph of f is shown in the figure above. Let R be the region bounded by the graph of f and the x -axis.

- Find the area of R .
- Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.
- The portion of the region R for $1 \leq y \leq 2$ is revolved about the x -axis to form a solid. Find the volume of the solid.

Part A

There are closed form expressions for the antiderivatives. The response may give an exact value for the area, which is $\frac{4}{3} + \frac{1}{2}(\pi^2 - 8)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ definite integral for $-2 \leq x < 0$
- ☐ definite integral for $0 \leq x \leq \frac{\pi^2}{4}$

8-5-7

☐ answer**Solution:**

$$\text{Area} = \int_{-2}^0 \frac{1}{2}(x+2)^2 dx + \int_0^{\frac{\pi^2}{4}} (2 - 2 \sin \sqrt{x}) dx = 2.268$$

Part B

At most 1 out of 2 points earned if there is one error in the integrand OR 1 error with limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ integrands
☐ expression for total volume

Solution:

$$\text{Volume} = \int_{-2}^0 \left(\frac{1}{2}(x+2)^2 \right)^2 dx + \int_0^{\frac{\pi^2}{4}} (2 - 2 \sin \sqrt{x})^2 dx$$

Part C

To be eligible for the fourth point, the response must have earned the first two points.

Special case: A response with the expression

$$\pi \int_{-0.585786}^0 \left(\frac{1}{2}(x+2)^2 \right)^2 dx + \pi \int_0^{0.274156} (2 - 2 \sin \sqrt{x})^2 dx = 5.688 \text{ (or } 5.687)$$

earns the third and fourth points.

There are closed form expressions for the antiderivatives. The response may give an exact values for the limits and volume. The limits are

$x_1 = \sqrt{2} - 2$ and $x_2 = \frac{\pi^2}{36}$. An unsimplified expression for the volume is given by

$$\frac{\pi}{20} \left(2^5 - (\sqrt{2})^5 \right) + \pi \left(\sqrt{2} - 2 \right) + \frac{\pi}{36} \left(-270 + 42\sqrt{3}\pi + 5\pi^2 \right).$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ integrand for one region

8-5-7

- ☐ integrand for the remaining region
- ☐ limits
- ☐ constant and answer

Solution:

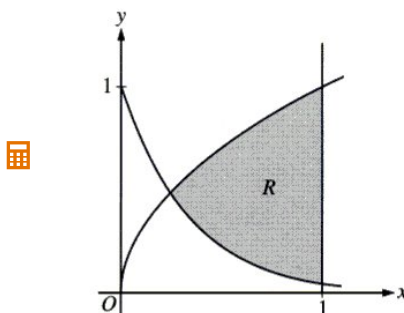
$$1 = \frac{1}{2}(x_1 + 2)^2 \Rightarrow x_1 = -0.585786$$

$$1 = 2 - 2 \sin \sqrt{x_2} \Rightarrow x_2 = 0.274156$$

$$\begin{aligned} \text{Volume} &= \pi \int_{x_1}^0 \left(\left(\frac{1}{2}(x+2)^2 \right)^2 - 1^2 \right) dx + \pi \int_0^{x_2} \left((2 - 2 \sin \sqrt{x})^2 - 1^2 \right) dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

8-5-7

17.



Let R be the shaded region bounded by the graphs of $y = \sqrt{x}$ and $y = e^{-3x}$ and the vertical line $x = 1$, as shown in the figure above.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is revolved about the horizontal line $y = 1$.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 5 times the length of its base in region R . Find the volume of this solid.

Part A

1 point is earned for the correct integrand

$$\text{Area} = \int_T^1 (\sqrt{x} - e^{-3x}) dx$$

1 point is earned for the correct answer

$$= 0.442 \text{ or } 0.443$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct integrand

$$\text{Area} = \int_T^1 (\sqrt{x} - e^{-3x}) dx$$

1 point is earned for the correct answer

$$= 0.442 \text{ or } 0.443$$

1 point is earned for correct limits in an integral in (a), (b), (c).

8-5-7

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

Part B

2 points are earned for the correct integrand

<-1> reversal

<-1> error with constant

<-1> omits 1 in one radius

<-2> other errors

$$\text{Volume} = \pi \int_1^T ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct integrand

<-1> reversal

<-1> error with constant

<-1> omits 1 in one radius

<-2> other errors

$$\text{Volume} = \pi \int_1^T ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

8-5-7

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

Part C

2 points are earned for the correct integrand

<-1> incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct integrand

<-1> incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

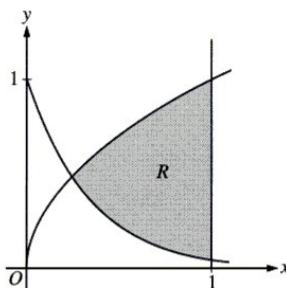
1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

8-5-7

18.



Let R be the shaded region bounded by the graphs of $y = \sqrt{x}$ and $y = e^{-3x}$ and the vertical line $x = 1$, as shown in the figure above.

 The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 5 times the length of its base in region R . Find the volume of this solid.

Part C

2 points are earned for the correct integrand

< -1 > incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct integrand

< -1 > incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

Overall

1 point is earned for for the correct limits in an integral in (a), (b), or (c)

Point of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

✓

0	1
---	---

8-5-7

The student response earns one of the following points:

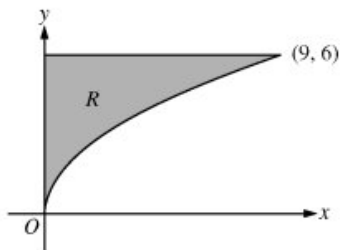
1 point is earned for for the correct limits in an integral in (a), (b), or (c)

Point of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

8-5-7

19.



Let R be the region in the first quadrant bounded by the graph of $y = 2\sqrt{x}$, the horizontal line $y = 6$, and the y -axis, as shown in the figure above.

(a) Find the area of R .

(b) Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 7$.

(c) Region R is the base of a solid. For each y , where $0 \leq y \leq 6$, the cross section of the solid taken perpendicular to the y -axis is a rectangle whose height is 3 times the length of its base in region R . Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{3/2} \right) \Big|_{x=0}^{x=9} = 18$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{3/2} \right) \Big|_{x=0}^{x=9} = 18$$

Part B

2 points are earned for integrand

1 point is earned for limits and constant

8-5-7

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$

Part C

2 points are earned for integrand

1 point is earned for the answer

$$\text{Solving } y = 2\sqrt{x} \text{ for } x \text{ yields } x = \frac{y^2}{4}.$$

$$\text{Each rectangular cross section has area } \left(3\frac{y^2}{4} \right) \left(\frac{y^2}{4} \right) = \frac{3}{16}y^4.$$

$$\text{Volume} = \int_0^6 \frac{3}{16}y^4 dy$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

$$\text{Solving } y = 2\sqrt{x} \text{ for } x \text{ yields } x = \frac{y^2}{4}.$$

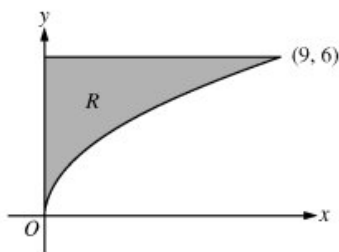
$$\text{Each rectangular cross section has area } \left(3\frac{y^2}{4} \right) \left(\frac{y^2}{4} \right) = \frac{3}{16}y^4.$$

8-5-7

$$\text{Volume} = \int_0^6 \frac{3}{16} y^4 dy$$

8-5-7

20.



Let R be the region in the first quadrant bounded by the graph of $y = 2\sqrt{x}$, the horizontal line $y = 6$, and the y -axis, as shown in the figure above.

Region R is the base of a solid. For each y , where $0 \leq y \leq 6$ the cross section of the solid taken perpendicular to the y -axis is a rectangle whose height is 3 times the length of its base in region R . Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part C

2 point is earned for integral

1 point is earned for the answer

Solving $y = 2\sqrt{x}$ for x yields $x = \frac{y^2}{4}$

Each rectangular cross section area $\left(3\frac{y^2}{4}\right)\left(\frac{y^2}{4}\right) = \frac{3}{16}y^4$

$$\text{Volume} = \int_0^6 \frac{3}{16}y^4 dy$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 point is earned for integral

1 point is earned for the answer

Solving $y = 2\sqrt{x}$ for x yields $x = \frac{y^2}{4}$

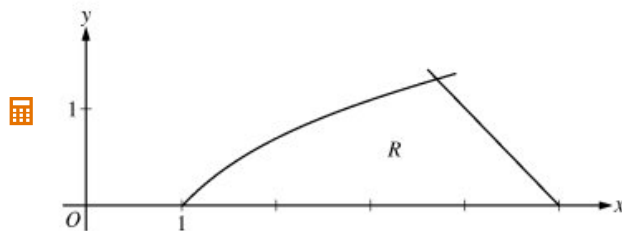
Each rectangular cross section area $\left(3\frac{y^2}{4}\right)\left(\frac{y^2}{4}\right) = \frac{3}{16}y^4$

8-5-7

$$Volume = \int_0^6 \frac{3}{16} y^4 \, dy$$

8-5-7

21.



Let R be the region in the first quadrant bounded by the x -axis and the graphs of $y = \ln x$ and $y = 5 - x$, as shown in the figure above.

(a) Find the area of R .

(b) Region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

(c) The horizontal line $y = k$ divides R into two regions of equal area. Write, but do not solve, an equation involving one or more integrals whose solution gives the value of k .

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\begin{aligned} \text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

OR

$$\begin{aligned} \text{Area} &= \int_1^A \ln x \, dx + \int_A^5 (5 - x) \, dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

8-5-7

$$\begin{aligned}\text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 \text{ (or } 2.985)\end{aligned}$$

OR

$$\begin{aligned}\text{Area} &= \int_1^A \ln x \, dx + \int_A^5 (5 - x) dx \\ &= 2.986 \text{ (or } 2.985)\end{aligned}$$

Part B

2 points are earned for the integrands

1 point is earned for expression for total volume

$$\text{Volume} = \int_1^A (\ln x)^2 dx + \int_A^5 (5 - x)^2 dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrands

1 point is earned for expression for total volume

$$\text{Volume} = \int_1^A (\ln x)^2 dx + \int_A^5 (5 - x)^2 dx$$

Part C

1 point is earned for the integrand

1 points are earned the limit

1 point is earned for the equation

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \left(\text{or } \frac{1}{2} \cdot 2.985 \right)$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

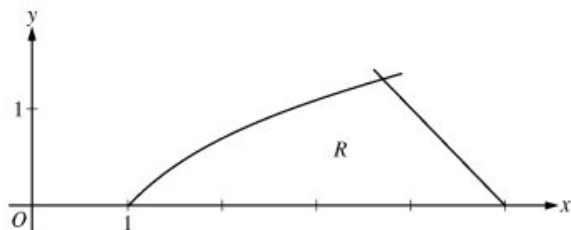
1 point is earned for the integrand

8-5-7

1 point is earned the limit

1 point is earned for the equation

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \left(\text{or } \frac{1}{2} \cdot 2.985 \right)$$



Let R be the region in the first quadrant bounded by the x -axis and the graphs of $y = \ln x$ and $y = 5 - x$, as shown in the figure above.

22.  Region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part B

The student response earns none of the following points:

2 points are earned for the integrands

1 point is earned for expression for total volume

$$Volume = \int_1^A (\ln x)^2 + \int_A^5 (5 - x)^2 dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrands

1 point is earned for expression for total volume

$$Volume = \int_1^A (\ln x)^2 + \int_A^5 (5 - x)^2 dx$$

8-5-7

23.  The horizontal line $y = k$ divides R into two regions of equal area. Write, but do not solve, an equation involving one or more integrals whose solution gives the value of k .

Part C

1 point is earned for the integrand

1 points are earned the limit

1 point is earned for the equation

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \quad \left(\text{or } \frac{1}{2} \cdot 2.985\right)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

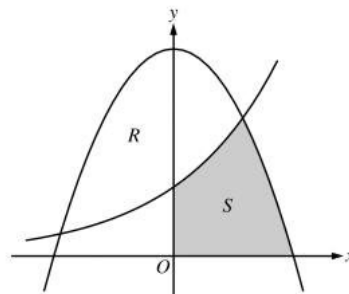
1 points are earned the limit

1 point is earned for the equation

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \quad \left(\text{or } \frac{1}{2} \cdot 2.985\right)$$

8-5-7

24.



Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.

- Find the area of S .
- Find the volume of the solid generated when R is rotated about the horizontal line $y = -1$.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with one leg across the base of the solid. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for integrands

1 point is earned for limits

1 point is earned for the answer

$$3 - x^2 = 2^x \text{ when } x = -1.63658 \text{ and } x = 1$$

Let $a = -1.63658$

$$\begin{aligned} \text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrands

1 point is earned for limits

1 point is earned for the answer

$$3 - x^2 = 2^x \text{ when } x = -1.63658 \text{ and } x = 1$$

Let $a = -1.63658$

8-5-7

$$\begin{aligned}\text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240\end{aligned}$$

Part B

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_a^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106 \text{ or } 63.107\end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_a^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106 \text{ or } 63.107\end{aligned}$$

Part C

1 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$



0	1	2
---	---	---

The student response earns all of the following points:

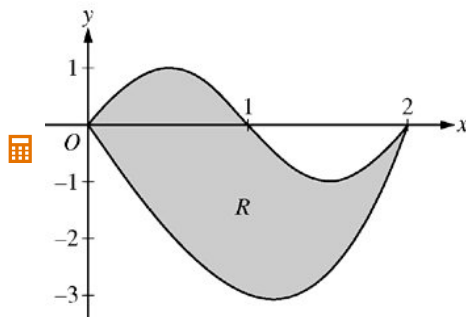
1 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$

8-5-7

25.



Let R be the region bounded by the graphs of $y = \sin(\pi x)$ and $y = x^3 - 4x$, as shown in the figure above.

(a) Find the area of R .

(b) The horizontal line $y = -2$ splits the region R into two parts. Write, but do not evaluate, an integral expression for the area of the part of R that is below this horizontal line.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.

(d) The region R models the surface of a small pond. At all points in R at a distance x from the y -axis, the depth of the water is given by $h(x) = 3 - x$. Find the volume of water in the pond.

Part A

The response can earn up to 3 points:

1 point is earned for the correct limits

$$\sin(\pi x) = x^3 - 4x \text{ at } x = 0 \text{ and } x = 2$$

1 point is earned for the correct integrand

1 point is earned for the correct answer

$$\text{Area} = \int_0^2 (\sin(\pi x) - (x^3 - 4x)) dx = 4$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point is earned for the correct limits

$$\sin(\pi x) = x^3 - 4x \text{ at } x = 0 \text{ and } x = 2$$

1 point is earned for the correct integrand

1 point is earned for the correct answer

$$\text{Area} = \int_0^2 (\sin(\pi x) - (x^3 - 4x)) dx = 4$$

Part B

The response can earn up to 2 points:

8-5-7

1 point is earned for the correct limits

$$x^3 - 4x = -2 \text{ at } r = 0.5391889 \text{ and } s = 1.6751309$$

1 point is earned for the correct integrand

The area of the stated region is $\int_r^s (-2 - (x^3 - 4x))dx$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the correct limits

$$x^3 - 4x = -2 \text{ at } r = 0.5391889 \text{ and } s = 1.6751309$$

1 point is earned for the correct integrand

The area of the stated region is $\int_r^s (-2 - (x^3 - 4x))dx$

Part C

The response can earn up to 2 points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (\sin(\pi x) - (x^3 - 4x))^2 dx = 9.978$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (\sin(\pi x) - (x^3 - 4x))^2 dx = 9.978$$

Part D

The response can earn up to 2 points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (3 - x) (\sin(\pi x) - (x^3 - 4x)) dx = 8.369 \text{ or } 8.370$$

8-5-7

✓

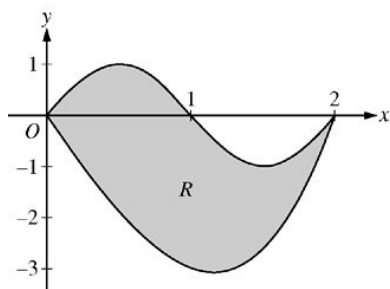
0	1	2
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The response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the correct answer

$$\text{Volume} = \int_0^2 (3 - x) (\sin(\pi x) - (x^3 - 4x)) dx = 8.369 \text{ or } 8.370$$



Let R be the region bounded by the graphs of $y = \sin(\pi x)$ and $y = x^3 - 4x$, as shown in the figure above.

26.  The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.

Part C

The response can earn up to 2 points:

1 point: For the integrand

1 point: For the correct answer

$$\text{Volume} = \int_0^2 (\sin(\pi x) - (x^3 - 4x))^2 dx = 9.978$$

✓

0	1	2
---	---	---

The response earns both of the following points:

1 point: For the integrand

1 point: For the correct answer

$$\text{Volume} = \int_0^2 (\sin(\pi x) - (x^3 - 4x))^2 dx = 9.978$$

8-5-7

27.  The region R models the surface of a small pond. At all points in R at a distance x from the y -axis, the depth of the water is given by $h(x) = 3 - x$. Find the volume of water in the pond.
-

Part D

The response can earn up to 2 points:

1 point: For the integrand

1 point: For the correct answer

$$\text{Volume} = \int_0^2 (3 - x) (\sin(\pi x) - (x^3 - 4x)) dx = 8.369 \text{ or } 8.370$$



0	1	2
---	---	---

The response earns both of the following points:

1 point: For the integrand

1 point: For the correct answer

$$\text{Volume} = \int_0^2 (3 - x) (\sin(\pi x) - (x^3 - 4x)) dx = 8.369 \text{ or } 8.370$$

8-5-7

28. Let $x = ky^2 + 2$, where $k > 0$.

Let R be the region in the first quadrant bounded by the x -axis, the graph of $x = ky^2 + 2$, and the line $x = 4$. Write an integral expression for the area of the region R and show that this area decreases as k increases.

Part C

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2)) dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x - 2} dx$$

< -1 > error in limits

< -1 > considers twice the region

< -2 > error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

$$A = \frac{4\sqrt{2}}{3} k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3} k^{-1.5} < 0$$

for all $k > 0$

thus the area decreases as k increases



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2)) dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x - 2} dx$$

< -1 > error in limits

< -1 > considers twice the region

< -2 > error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

$$A = \frac{4\sqrt{2}}{3} k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3} k^{-1.5} < 0$$

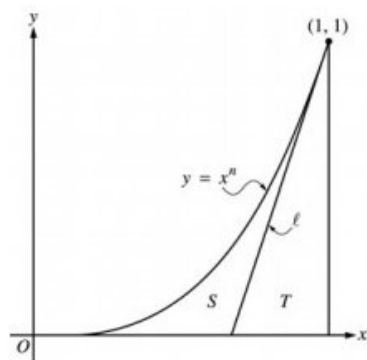
8-5-7

for all $k > 0$

thus the area decreases as k increases

8-5-7

29.



Let l be the line tangent to the graph of $y = x^n$ at the point $(1, 1)$, where $n > 1$, as shown above.

Let S be the region bounded by the graph of $y = x^n$, the line l , and the x -axis. Express the area of S in terms of n and determine the value of n that maximizes the area of S .

Part C

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

$$\begin{aligned}
 \text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\
 &= \frac{1}{n+1} - \frac{1}{2n} \\
 \frac{d}{dn} \text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\
 2n^2 &= (n+1)^2 \\
 \sqrt{2}n &= (n+1) \\
 n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}
 \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

8-5-7

$$\text{Area}(S) = \int_0^1 x^n dx - \text{Area}(T)$$

$$= \frac{1}{n+1} - \frac{1}{2n}$$

$$\frac{d}{dn} \text{Area}(S) = -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0$$

$$2n^2 = (n+1)^2$$

$$\sqrt{2}n = (n+1)$$

$$n = \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}$$

8-5-7

30. Consider the curve $y^2=4+x$ and chord AB joining the points $A(-4,0)$ and $B(0,2)$ on the curve.
Find the area of the region R enclosed by the curve and the chord AB .

Part B

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_{-4}^0 \left[(\sqrt{4+x} - (\frac{1}{2}x + 2)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3} \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_0^2 [(2y-4) - (y^2-4)] dy &= y^2 - \frac{y^3}{3} \Big|_0^2 \\ &= 4 - \frac{8}{3} = \frac{4}{3} \end{aligned}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

8-5-7

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_{-4}^0 \left[(\sqrt{4+x} - (\frac{1}{2}x + 2)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3} \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_0^2 [(2y-4) - (y^2-4)] dy &= y^2 - \frac{y^3}{3} \Big|_0^2 \\ &= 4 - \frac{8}{3} = \frac{4}{3} \end{aligned}$$

Method 3:

1 point is earned for the correct bounds and constant

8-5-7

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

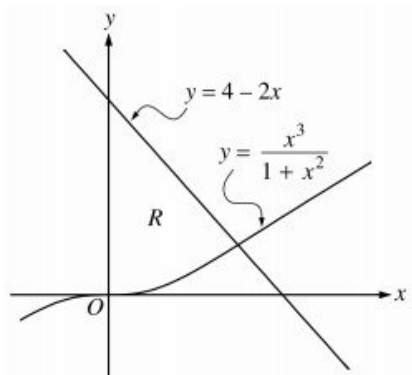
$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

8-5-7

31.



Let R be the region bounded by the y -axis and the graphs of $y = \frac{x^3}{1+x^2}$ and $y = 4 - 2x$, as shown in the figure above.



The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.

Part C

2 points are earned for the integrand $< -1 >$ each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \int_0^A \left((4 - 2x) - \frac{x^3}{1 + x^2} \right)^2 dx \\ &= 8.997 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand $< -1 >$ each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \int_0^A \left((4 - 2x) - \frac{x^3}{1 + x^2} \right)^2 dx \\ &= 8.997 \end{aligned}$$

32. The base of a solid is the region in the first quadrant bounded by the y -axis, the graph of $y = \tan^{-1} x$, the horizontal line $y = 3$ and the vertical line $x = 1$. For this solid, each cross section perpendicular to the x -axis is a square. What is the volume of the solid?

(A) 2.561

(B) 6.612

(C) 8.046

(D) 8.755

(E) 20.773



8-5-7

33.  Let R be the region in the first quadrant bounded below by the graph of $y = x^2$ and above by the graph of $y = \sqrt{x}$. R is the base of a solid whose cross sections perpendicular to the x -axis are squares. What is the volume of the solid?

(A) 0.129



(B) 0.300

(C) 0.333

(D) 0.700

(E) 1.271

34.  Let R be the region in the first quadrant bounded above by the graph of $y = \ln(3 - x)$, for $0 \leq x \leq 2$. R is the base of a solid for which each cross section perpendicular to the x -axis is a square. What is the volume of the solid?

(A) 0.442

(B) 1.029



(C) 1.296

(D) 3.233

(E) 4.071

35.  Let R be the region in the first quadrant bounded by the y -axis, the x -axis, the graph of $y = e^{-x^2/2}$, and the line $x=3$. The region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is a square. What is the volume of the solid?

(A) 0.886



(B) 0.906

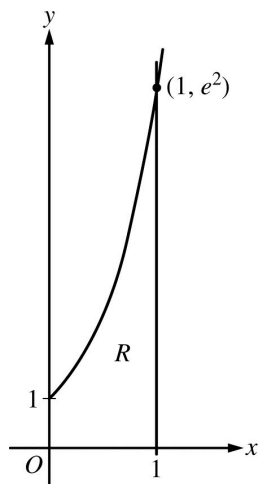
(C) 1.078

(D) 1.245

(E) 2.784

8-5-7

36.



Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.

Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integral that gives the volume of the solid.

Part C

The response can earn up to 4 points:

2 points: For the integrand

1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$

✓

0	1	2	3	4
---	---	---	---	---

The response earns all four of the following points:

2 points: For the integrand

1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$

8-5-7

37. Let R be the region in the first quadrant under the graph of $y = \frac{1}{\sqrt{x}}$ for $4 \leq x \leq 9$.

Find the volume of the solid whose base is the region R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part C

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

1 point is earned for the correct answer

0/1 if integrand is incorrect

$$\text{Volume} = \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx$$

$$= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4}$$

(or 0.811)



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

1 point is earned for the correct answer

0/1 if integrand is incorrect

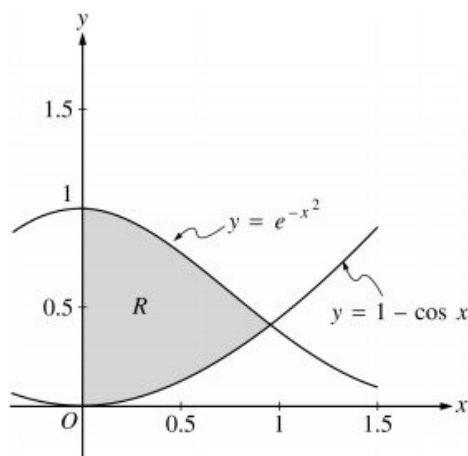
$$\text{Volume} = \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx$$

$$= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4}$$

(or 0.811)

8-5-7

38.



Let R be the shaded region in the first quadrant enclosed by the graphs of $y = e^{-x^2}$, $y = 1 - \cos x$, and the y -axis, as shown in the figure above.

 The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.

Part C

2 points are earned for the integrand

< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \int_0^1 (e^{-x^2} - (1 - \cos x)^2) dx \\ &= 0.461 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

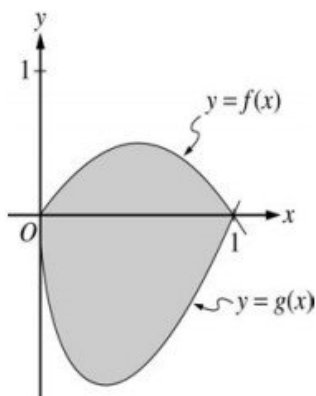
< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \int_0^1 (e^{-x^2} - (1 - \cos x)^2) dx \\ &= 0.461 \end{aligned}$$

8-5-7

39.



Let f and g be the functions given by $f(x) = 2x(1 - x)$ and $g(x) = 3(x - 1)\sqrt{x}$ for $0 \leq x \leq 1$. The graphs of f and g are shown in the figure above.

Let h be the function given by $h(x) = kx(1 - x)$ for x . For each $k > 0$, the region (not shown) enclosed by the graphs of h and g is the base of a solid with square cross sections perpendicular to the x -axis. There is a value of k for which the volume of this solid is equal to 15. Write, but do not solve, an equation involving an integral expression that could be used to find the value of k .

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_0^1 (h(x) - g(x))^2 dx$$

$$\int_0^1 (kx(1 - x) - 3(x - 1)\sqrt{x})^2 dx = 15$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_0^1 (h(x) - g(x))^2 dx$$

$$\int_0^1 (kx(1 - x) - 3(x - 1)\sqrt{x})^2 dx = 15$$

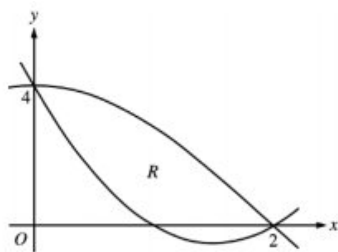
40. The base of a solid is the region in the first quadrant enclosed by the parabola $y = 4x^2$, the line $x = 1$, and the x -axis. Each plane section of the solid perpendicular to the x -axis is a square. The volume of the solid is

- (A) $4\pi/3$
 (B) $16\pi/5$
 (C) $4/3$
 (D) $16/5$
 (E) $64/5$

✓

8-5-7

41.



Let $f(x) = 2x^2 - 6x + 4$ and $g(x) = 4\cos(1/4 \pi x)$. Let R be the region bounded by the graphs of f and g , as shown in the figure above.

The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part C

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \int_0^2 [g(x) - f(x)]^2 dx$$

$$\int_0^2 [4\cos(\frac{\pi}{4}x) - (2x^2 - 6x + 4)]^2 dx$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \int_0^2 [g(x) - f(x)]^2 dx$$

$$\int_0^2 [4\cos(\frac{\pi}{4}x) - (2x^2 - 6x + 4)]^2 dx$$

8-5-7

42. Let R be the region in the first quadrant bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$.

The region R is the base of a solid. For this solid, the cross sections perpendicular to the y -axis are squares. Find the volume of this solid.

Part C

1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$



0	1	2
---	---	---

The student response earns all of the following points:

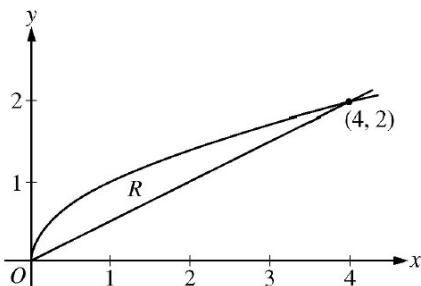
1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$

8-5-7

43.



Let R be the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$, as shown in the figure above.

The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are squares. Find the volume of this solid.

Part B

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Volume} = \int_0^4 (\sqrt{x} - \frac{x}{2})^2 dx = \int_0^4 (x - x^{\frac{3}{2}} + \frac{x^2}{4}) dx$$

$$= \frac{x^2}{2} - \frac{2x^{\frac{5}{2}}}{5} + \frac{x^3}{12} \bigg|_{x=0}^{x=4} = \frac{8}{15}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the antiderivative

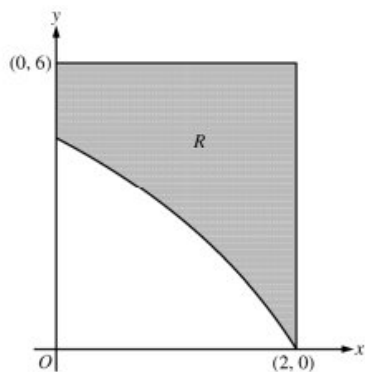
1 point is earned for the answer

$$\text{Volume} = \int_0^4 (\sqrt{x} - \frac{x}{2})^2 dx = \int_0^4 (x - x^{\frac{3}{2}} + \frac{x^2}{4}) dx$$

$$= \frac{x^2}{2} - \frac{2x^{\frac{5}{2}}}{5} + \frac{x^3}{12} \bigg|_{x=0}^{x=4} = \frac{8}{15}$$

8-5-7

44.



In the figure above, R is the shaded region in the first quadrant bounded by the graph of $y=4\ln(3-x)$, the horizontal line $y=6$, and the vertical line $x=2$.

 The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of the solid.

Part C

2 points are earned for the integrand

1 point is earned for the answer

1 point is earned for correct limits in an integral in (a), (b), or (c)

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

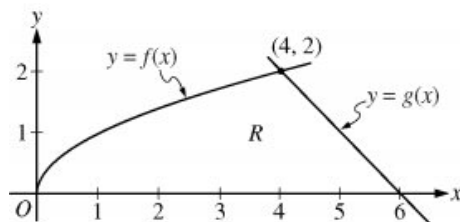
1 point is earned for the answer

1 point is earned for correct limits in an integral in (a), (b), or (c)

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$

8-5-7

45.



The functions f and g are given by $f(x) = \sqrt{x}$ and $g(x) = 6 - x$. Let R be the region bounded by the x -axis and the graphs of f and g , as shown in the figure above.

The region R is the base of a solid. For each y , where $0 \leq y \leq 2$, the cross section of the solid taken perpendicular to the y -axis is a rectangle whose base lies in R and whose height is $2y$. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part B

2 point is earned for integrand

1 point is earned for the answer

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$



0	1	2	3
---	---	---	---

The student response earns none of the following points:

2 point is earned for integrand

1 point is earned for the answer

$$y = \sqrt{x} \Rightarrow x = y^2$$

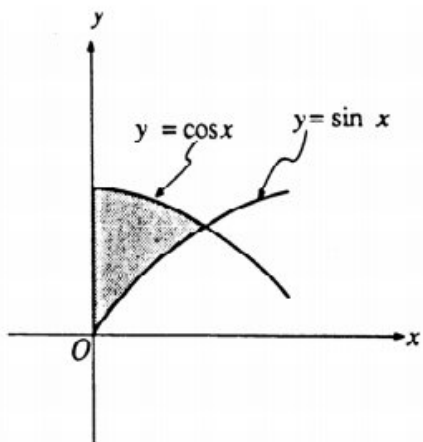
$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$

8-5-7

46.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = \sin x$ and $y = \cos x$, as shown in the figure above.

Find the volume of the solid whose base is R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part C

2 points are earned for integral

1 point is **deducted** for constant or limit error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \int_0^{\pi/4} (\cos x - \sin x)^2 dx \\
 &= \int_0^{\pi/4} 1 - 2 \sin x \cos x dx \\
 &= (x - \sin^2 x) \Big|_0^{\pi/4} \\
 &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\
 &= \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for integral

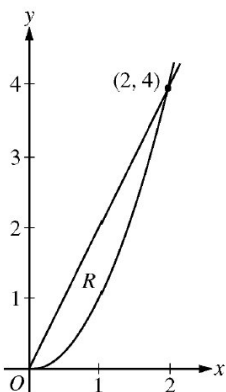
1 point is **deducted** for constant or limit error

1 point is earned for antiderivative and evaluation

8-5-7

$$\begin{aligned}
 V &= \int_0^{\pi/4} (\cos x - \sin x)^2 dx \\
 &= \int_0^{\pi/4} 1 - 2 \sin x \cos x dx \\
 &= (x - \sin^2 x) \Big|_0^{\pi/4} \\
 &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\
 &= \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$

47.



Let R be the region in the first quadrant enclosed by the graphs of $y=2x$ and $y=x^2$, as shown in the figure above.

Another solid has the same base R . For this solid, the cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an integral expression for the volume of the solid.

Part C

2 points are earned for the integrand

1 point is earned for the limit

$$volume = \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the integrand

1 point is earned for the limit

$$volume = \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$$

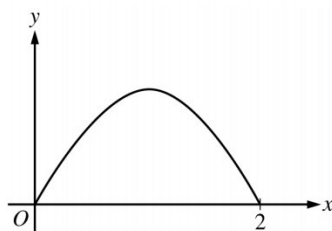
8-5-7

48. The region enclosed by the graphs of $y = x^2$ and $y = 2x$ is the base of a solid. For the solid, each cross section perpendicular to the y -axis is a rectangle whose height is 3 times its base in the xy -plane.

Which of the following expressions gives the volume of the solid?

- (A) $3 \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$ ✓
- (B) $3 \int_0^4 \left(\sqrt{y} + \frac{y}{2} \right)^2 dy$
- (C) $3 \int_0^2 (2x - x^2)^2 dx$
- (D) $3 \int_0^2 (2x + x^2)^2 dx$

49.



The base of a solid is the region bounded by a portion of the graph of $y = \sin\left(\frac{\pi}{2}x\right)$ and the x -axis, as shown in the figure above. For the solid, each cross section perpendicular to the x -axis is a rectangle of height 3.

Which of the following expressions gives the volume of the solid?

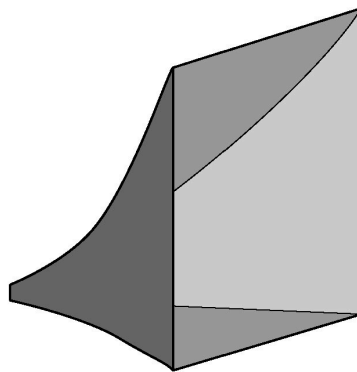
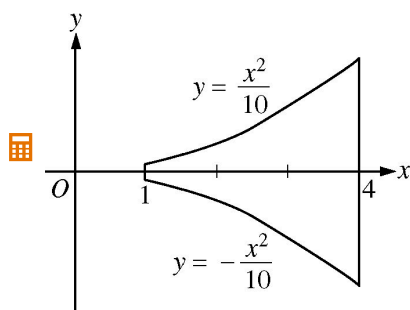
- (A) $\int_0^2 3 \sin\left(\frac{\pi}{2}x\right) dx$ ✓
- (B) $\int_0^2 3 \sin^2\left(\frac{\pi}{2}x\right) dx$
- (C) $\int_0^2 3\pi \sin\left(\frac{\pi}{2}x\right) dx$
- (D) $\int_0^2 3\pi \sin^2\left(\frac{\pi}{2}x\right) dx$

50. The base of a solid is the region bounded by the x -axis and the graph of $y = \sqrt{1 - x^2}$. For the solid, each cross section perpendicular to the x -axis is a square. What is the volume of the solid?

- (A) $\frac{2}{3}$
- (B) $\frac{4}{3}$ ✓
- (C) 2
- (D) $\frac{2\pi}{3}$
- (E) $\frac{4\pi}{3}$

8-5-7

51.



The base of a loudspeaker is determined by the two curves $y = \frac{x^2}{10}$ and $y = -\frac{x^2}{10}$ for as shown in the figure above. For this loudspeaker, the cross sections perpendicular to the x -axis are squares. What is the volume of the loudspeaker, in cubic units?

- (A) 2.046
(B) 4.092
(C) 4.200
(D) 8.184
(E) 25.711



8-5-7

52. Let $x = ky^2 + 2$, where $k > 0$.

(a) Show that for all $k > 0$, the point $\left(4, \sqrt{\frac{2}{k}}\right)$ is on the graph of $x = ky^2 + 2$.

(b) Show that for all $k > 0$, the tangent line to the graph of $x = ky^2 + 2$ at the point $\left(4, \sqrt{\frac{2}{k}}\right)$ passes through the origin.

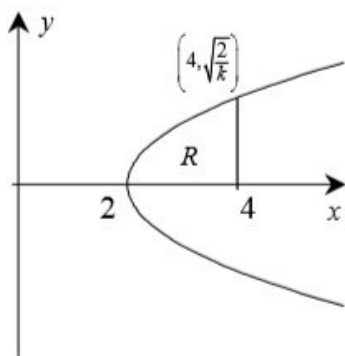
(c) Let R be the region in the first quadrant bounded by the x -axis, the graph of $x = ky^2 + 2$, and the line $x = 4$. Write an integral expression for the area of the region R and show that this area decreases as k increases.

Part A

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$$4 = k\left(\frac{2}{k}\right) + 2$$

$$4 = 4$$



0

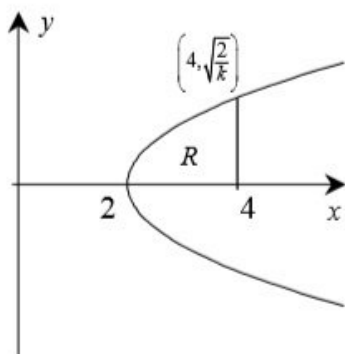
1

The student response earns all of the following points:

1 point is earned for correctly substituting $x = 4$ and $y = \sqrt{\frac{2}{k}}$

$$4 = k\left(\frac{2}{k}\right) + 2$$

$$4 = 4$$



Part B

8-5-7

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$$x = ky^2 + 2$$

$$1 = 2ky \frac{dy}{dx}$$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$$\frac{dy}{dx} \Big|_{y=\sqrt{2/k}} = \frac{1}{2\sqrt{2k}}$$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{2/k}\right)$ and (0,0)

the tangent line is

$$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x - 4)$$

$$y = \frac{1}{2\sqrt{2k}}x \text{ which contains } (0,0)$$

or

$$\text{slope of line through } (0,0) \text{ and } \left(4, \sqrt{2/k}\right) \text{ is } \frac{\sqrt{2/k}}{4} = \frac{1}{2\sqrt{2k}}$$

which is the same as the slope of the tangent line

1 point is earned for correctly observing (0,0) is a solution or observing the slopes are equal



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct equation with $\frac{dy}{dx}$ or $\frac{dx}{dy}$

$$x = ky^2 + 2$$

$$1 = 2ky \frac{dy}{dx}$$

1 point is earned for the correct evaluation of $\frac{dy}{dx}$ or $\frac{dx}{dy}$ at $\left(4, \sqrt{\frac{2}{k}}\right)$

$$\frac{dy}{dx} \Big|_{y=\sqrt{2/k}} = \frac{1}{2\sqrt{2k}}$$

1 point is earned for the correct equation of tangent line or slope of line using $\left(4, \sqrt{2/k}\right)$ and (0,0)

the tangent line is

$$y - \sqrt{\frac{2}{k}} = \frac{1}{2\sqrt{2k}}(x - 4)$$

$$y = \frac{1}{2\sqrt{2k}}x \text{ which contains } (0,0)$$

8-5-7

or

slope of line through (0,0) and $\left(4, \sqrt{2/k}\right)$ is $\frac{\sqrt{2/k}}{4} = \frac{1}{2\sqrt{2k}}$

which is the same as the slope of the tangent line

1 point is earned for correctly observing (0,0) is a solution or observing the slopes are equal

Part C

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2)) dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x-2} dx$$

<-1> error in limits

<-1> considers twice the region

<-2> error in integrand

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

$$A = \frac{4\sqrt{2}}{3} k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3} k^{-1.5} < 0 \text{ for all } k > 0$$

thus the area decreases as k increases



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

3 points are earned for correctly expressing the area as a definite integral

$$A = \int_0^{\sqrt{2/k}} (4 - (ky^2 + 2)) dy$$

or

$$A = \frac{1}{\sqrt{k}} \int_2^4 \sqrt{x-2} dx$$

<-1> error in limits

<-1> considers twice the region

<-2> error in integrand

8-5-7

1 point is earned for correctly showing that A decreases using algebra, geometry, or calculus

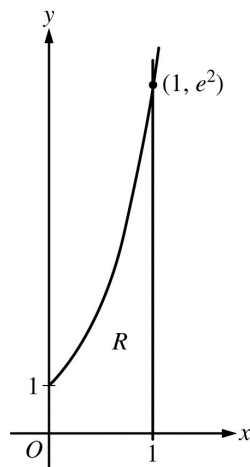
$$A = \frac{4\sqrt{2}}{3}k^{-0.5}$$

$$\frac{dA}{dk} = -\frac{2\sqrt{2}}{3}k^{-1.5} < 0 \text{ for all } k > 0$$

thus the area decreases as k increases

8-5-7

53.



Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.

(a) Write an equation for the line tangent to the graph of f at $x = 1$.

(b) Find the area of R .

(c) Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part A

The response can earn up to 2 points:

1 point: For $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.



0	1	2
---	---	---

The response earns all of the following points:

1 point: For $f'(1)$

$$f(1) = e^2$$

$$f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$$

1 point: For the correct answer

An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.

8-5-7

Part B

The response can earn up to 3 points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2} (e^2 - 1)$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the integral

1 point: For the antiderivative

1 point: For the correct answer

$$\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2} (e^2 - 1)$$

Part C

The response can earn up to 4 points:

2 points: For the integrand

1 point: For correct limits

1 point: For the correct answer

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y \right)^2 dy$$



0	1	2	3	4
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The response earns all of the following points:

2 points: For the integrand

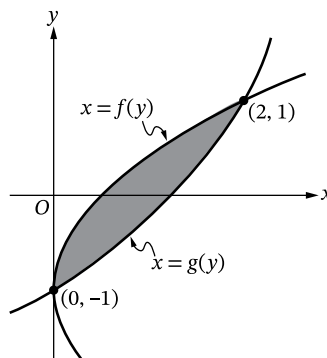
1 point: For correct limits

1 point: For the correct answer

8-5-7

$$\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y\right)^2 dy$$

54. Let R be the region bounded on the left by the graph of $x = f(y)$ and on the right by the graph of $x = g(y)$, as shaded in the figure.



Which of the following gives the area of R ?

- (A) $\int_{-1}^1 (f(y) - g(y)) dy$
- (B) $\int_{-1}^1 (g(y) - f(y)) dy$
- (C) $\int_0^2 (f(y) - g(y)) dy$
- (D) $\int_0^2 (g(y) - f(y)) dy$



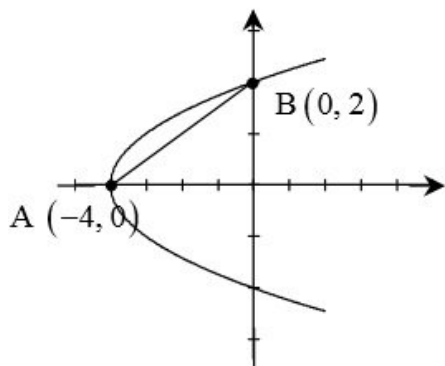
Answer B

Correct. Since the curves are given as functions of y , the region can be sliced along the y -axis so that the area is expressed as a definite integral in terms of y . The right curve is given by $g(y)$ and the left curve is given by $f(y)$. The curves intersect at $y = -1$ and $y = 1$. Therefore the area is given by the expression $\int_{-1}^1 (g(y) - f(y)) dy$.

8-5-7

55. Consider the curve $y^2 = 4 + x$ and chord AB joining the points $A(-4, 0)$ and $B(0, 2)$ on the curve.
- Find the x - and y -coordinates of the point on the curve where the tangent line is parallel to chord AB .
 - Find the area of the region R enclosed by the curve and the chord AB .
 - Find the volume of the solid generated when the region R , defined in part (b), is revolved about the x -axis.

Part A



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4 + x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4 + x}}; \frac{1}{2\sqrt{4 + x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

$$\text{and } y = 1, x = -3$$

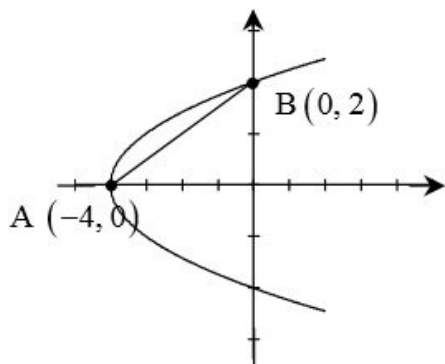
Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

8-5-7



0	1	2	3
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The student response earns all of the following points:



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4+x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4+x}}; \frac{1}{2\sqrt{4+x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

$$\text{and } y = 1, x = -3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part B

Method 1:

8-5-7

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned}\int_{-4}^0 \left[(\sqrt{4+x} - \left(\frac{1}{2}x + 2\right)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3}\end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned}\int_0^2 [(2y - 4) - (y^2 - 4)] dy &= y^2 - \frac{y^3}{3} \Big|_0^2 \\ &= 4 - \frac{8}{3} = \frac{4}{3}\end{aligned}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

8-5-7

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_{-4}^0 \left[(\sqrt{4+x} - \left(\frac{1}{2}x + 2\right)) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3} \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_0^2 [(2y-4) - (y^2-4)] dy &= y^2 - \frac{y^3}{3} \Big|_0^2 \\ &= 4 - \frac{8}{3} = \frac{4}{3} \end{aligned}$$

8-5-7Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part CMethod 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \end{aligned}$$

$$= \pi(8 - 16/3) = 8\pi/3 \approx 8.378$$

Method 2:

8-5-7

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 (\sqrt{4+x})^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

$$\text{Volume of cone} = \frac{1}{3} \pi (2)^2 (4) = 16\pi/3$$

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

8-5-7



0	1	2	3
---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \end{aligned}$$

$$= \pi(8 - 16/3) = 8\pi/3 \approx 8.378$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

8-5-7

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 (\sqrt{4+x})^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

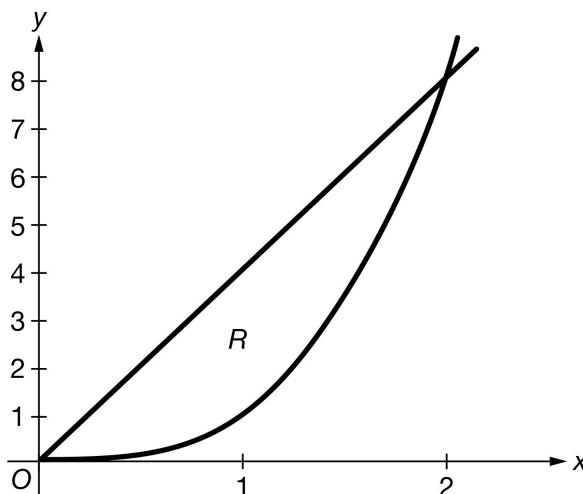
$$\text{Volume of cone} = \frac{1}{3} \pi (2)^2 (4) = 16\pi/3$$

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

8-5-7

56.



Let R be the region in the first quadrant bounded by the graphs of $y = x^3$ and $y = 4x$, as shown in the figure above. The region R is the base of a solid. For the solid, each cross section perpendicular to the y -axis is a square. What is the volume of the solid?

- (A) 0.992
 (B) 2.438
 (C) 5.461
 (D) 8.533



Answer B

Correct. Expressed as functions of y , the boundary curves are $x = \sqrt[3]{y}$ and $x = \frac{y}{4}$. Region R is bounded on the right by the graph of $x = \sqrt[3]{y}$ and on the left by the graph of $x = \frac{y}{4}$ for $0 \leq y \leq 8$. At each y , the cross section perpendicular to the y -axis has area $(\sqrt[3]{y} - \frac{y}{4})(\sqrt[3]{y} - \frac{y}{4})$. Therefore, the volume of the solid is given by the definite integral $\int_0^8 (\sqrt[3]{y} - \frac{y}{4})^2 dy$, which can be evaluated using the graphing calculator.

57. The base of a solid is the region in the first quadrant bounded by the graph of $y = \cos x$ and the x - and y -axes for $0 \leq x \leq \frac{\pi}{2}$. For the solid, each cross section perpendicular to the y -axis is a rectangle whose height is four times its width in the xy -plane. What is the volume of the solid?

- (A) 0.285
 (B) 3.142
 (C) 4.000
 (D) 4.566

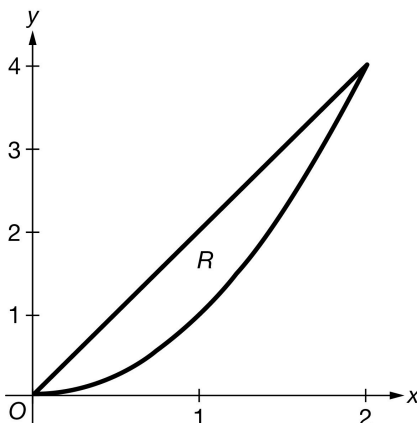


Answer D

Correct. Expressed as functions of y , the boundary curve $y = \cos x$ is $x = \cos^{-1}y$ for $0 \leq y \leq 1$. At each y , the cross section perpendicular to the y -axis has area $4(\cos^{-1}y)^2$. Therefore, the volume of the solid is given by the definite integral $\int_0^1 4(\cos^{-1}y)^2 dy$, which can be evaluated using the graphing calculator.

8-5-7

58.



Let R be the region in the first quadrant bounded by the graphs of $y = x^2$ and $y = 2x$, as shown in the figure above. The region R is the base of a solid. For the solid, each cross section perpendicular to the y -axis is a square. What is the volume of the solid?

- (A) 0.404
 (B) 0.533
 (C) 1.333
 (D) 2.667



Answer B

Correct. Expressed as functions of y , the boundary curves are $x = \sqrt{y}$ and $x = \frac{y}{2}$. Region R is bounded on the right by the graph of $x = \sqrt{y}$ and on the left by the graph of $x = \frac{y}{2}$ for $0 \leq y \leq 4$. At each y , the cross section perpendicular to the y -axis has area $(\sqrt{y} - \frac{y}{2})(\sqrt{y} - \frac{y}{2})$. Therefore, the volume of the solid is given by the definite integral $\int_0^4 (\sqrt{y} - \frac{y}{2})^2 dy$, which can be evaluated using the graphing calculator.

59. The base of a solid is the region in the first quadrant bounded by the graph of $y = 1 - \frac{x}{3}$ and the x - and y -axes. For the solid, each cross section perpendicular to the x -axis is a square. What is the volume of the solid?

- (A) $\frac{19}{27}$
 (B) $\frac{5}{6}$
 (C) 1
 (D) $\frac{3}{2}$



Answer C

Correct. For the region of the base, $0 \leq x \leq 3$. At each x , the cross section perpendicular to the x -axis has area $(1 - \frac{x}{3})(1 - \frac{x}{3})$.

Therefore, the volume of the solid is given by the definite integral $\int_0^3 (1 - \frac{x}{3})^2 dx$. This is evaluated as follows.

$$\int_0^3 \left(1 - \frac{x}{3}\right)^2 dx = \int_0^3 \left(1 - \frac{2}{3}x + \frac{x^2}{9}\right) dx = x - \frac{2}{3}\left(\frac{x^2}{2}\right) + \frac{1}{9}\left(\frac{x^3}{3}\right) \Big|_0^3 = \left(3 - \frac{9}{3} + \frac{27}{27}\right) - 0 = 1$$

8-5-7

60.  The base of a solid is the region in the first quadrant bounded by the graph of $y = 2 - \frac{2}{3}x$ and the x - and y -axes for $0 \leq x \leq 3$. For the solid, each cross section perpendicular to the y -axis is a rectangle whose height is five times its width in the xy -plane. What is the volume of the solid?
- (A) 1.2
(B) 15.0
(C) 20.0
(D) 30.0

Answer D

Correct. Expressed as functions of y , the boundary curve $y = 2 - \frac{2}{3}x$ is $x = \left(\frac{6-3y}{2}\right)$ for $0 \leq y \leq 2$. At each y , the cross section perpendicular to the y -axis has area $5\left(\frac{6-3y}{2}\right)^2$.

Therefore, the volume of the solid is given by the definite integral $\int_0^2 5\left(\frac{6-3y}{2}\right)^2 dy$, which can be evaluated using the graphing calculator.

61. The base of a solid is the region in the first quadrant bounded by the graph of $y = 2x - x^2$ and the x -axis. For the solid, each cross section perpendicular to the x -axis is a square. What is the volume of the solid?
- (A) $\frac{8}{15}$
(B) $\frac{2}{3}$
(C) $\frac{16}{15}$
(D) $\frac{4}{3}$

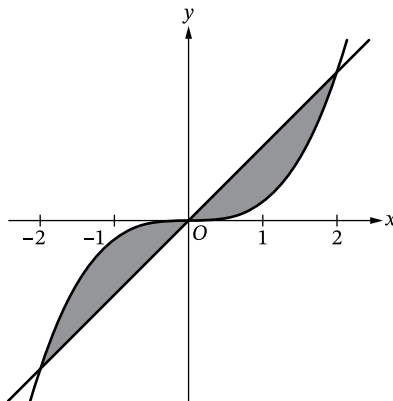
Answer C

Correct. For the region of the base, $0 \leq x \leq 2$. At each x , the cross section perpendicular to the x -axis has area $(2x - x^2)(2x - x^2)$. Therefore the volume of the solid is given by the definite integral $\int_0^2 (2x - x^2)^2 dx$. This is evaluated as follows.

$$\int_0^2 (2x - x^2)^2 dx = \int_0^2 (4x^2 - 4x^3 + x^4) dx = \left(\frac{4}{3}x^3 - x^4 + \frac{1}{5}x^5\right)\bigg|_0^2 = \left(\frac{32}{3} - 16 + \frac{32}{5}\right) - 0 = \frac{16}{15}$$

8-5-7

62. The regions enclosed by the graphs of $y = 4x$ and $y = x^3$ for $-2 \leq x \leq 2$ are shaded in the figure shown.



What is the total area of the shaded regions?

- (A) 0
(B) 4
(C) 8
(D) 12



Answer C

Correct. The two graphs intersect when $x^3 = 4x$: i.e., at the three values $x = -2$, $x = 0$, and $x = 2$. The total area can therefore be found as the sum of two definite integrals, as follows.

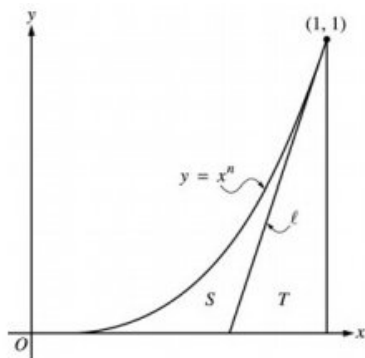
$$\int_{-2}^0 (x^3 - 4x) dx + \int_0^2 (4x - x^3) dx = \left(\frac{1}{4}x^4 - 2x^2 \right) \Big|_{-2}^0 + \left(2x^2 - \frac{1}{4}x^4 \right) \Big|_0^2 = 0 - (4 - 8) + (8 - 4) - 0 = 4 + 4 = 8.$$

Using the symmetry of the shaded region, the total area could also be found using just one definite integral, as follows.

$$2 \cdot \int_0^2 (4x - x^3) dx = 2 \cdot \left(2x^2 - \frac{x^4}{4} \right) \Big|_0^2 = 2 \cdot (8 - 4) = 8$$

8-5-7

63.



Let l be the line tangent to the graph of $y = x^n$ at the point $(1, 1)$, where $n > 1$, as shown above.

(a) Find $\int_0^1 x^n dx$ in terms of n .

(b) Let T be the triangular region bounded by l , the x -axis, and the line $x = 1$. Show that the area of T is $1/2n$.

(c) Let S be the region bounded by the graph of $y = x^n$, the line l , and the x -axis. Express the area of S in terms of n and determine the value of n that maximizes the area of S .

Part A

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$

✓

0	1	2
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The student response earns all of the following points:

1 point is earned for the antiderivative of x^n

1 point is earned for the answer

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1}$$

Part B

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

8-5-7

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if slope of line l is n

1 point is earned if base of T is $\frac{1}{n}$

1 point is earned if shows area is $\frac{1}{2n}$

Let b be the length of the base of triangle T .

$\frac{1}{b}$ is the slope of line l , which is n

$$\text{Area}(T) = \frac{1}{2}b(1) = \frac{1}{2n}$$

Part C

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

8-5-7

$$\begin{aligned}
 \text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\
 &= \frac{1}{n+1} - \frac{1}{2n} \\
 \frac{d}{dn} \text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\
 2n^2 &= (n+1)^2 \\
 \sqrt{2}n &= (n+1) \\
 n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}
 \end{aligned}$$

✓

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the area of S in terms of n

1 point is earned for the derivative

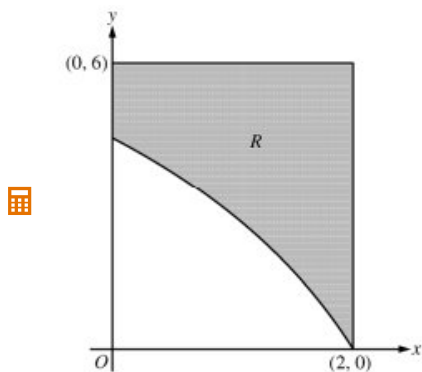
1 point is earned if sets derivative equal to 0

1 point is earned if solves for n

$$\begin{aligned}
 \text{Area}(S) &= \int_0^1 x^n dx - \text{Area}(T) \\
 &= \frac{1}{n+1} - \frac{1}{2n} \\
 \frac{d}{dn} \text{Area}(S) &= -\frac{1}{(n+1)^2} + \frac{1}{2n^2} = 0 \\
 2n^2 &= (n+1)^2 \\
 \sqrt{2}n &= (n+1) \\
 n &= \frac{1}{\sqrt{2}-1} = 1 + \sqrt{2}
 \end{aligned}$$

8-5-7

64.



In the figure above, R is the shaded region in the first quadrant bounded by the graph of $y = 4\ln(3 - x)$, the horizontal line $y = 6$, and the vertical line $x = 2$.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is revolved about the horizontal line $y = 8$.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of the solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x)) dx = 6.816 \text{ or } 6.817$$

1 point is earned for correct limits in an integral in (a), (b), or (c)



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x)) dx = 6.816 \text{ or } 6.817$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

Part B

2 points are earned for the integrand

1 point is earned for the answer

8-5-7

$$\pi \int_0^2 \left((8 - 4\ln(3 - x))^2 - (8 - 6)^2 \right) dx$$

$$= 168.179 \text{ or } 168.180$$

1 point is earned for correct limits in an integral in (a), (b), or (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\pi \int_0^2 \left((8 - 4\ln(3 - x))^2 - (8 - 6)^2 \right) dx$$

$$= 168.179 \text{ or } 168.180$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$

1 point is earned for correct limits in an integral in (a), (b), or (c)



0	1	2	3	4
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The student response earns all of the following points:

2 points are earned for the integrand

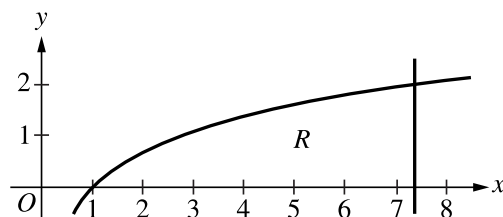
1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

8-5-7

65.



Let R be the region bounded by the graph of $y = \ln x$, the vertical line $x = e^2$, and the x -axis, as shown in the figure. Which of the following integrals are equal to the area of R ?

- I. $\int_1^{e^2} \ln x \, dx$
- II. $\int_0^2 e^y \, dy$
- III. $\int_0^2 (e^2 - e^y) \, dy$

- (A) I only
- (B) III only
- (C) I and II only
- (D) I and III only



Answer D

Correct. If the region is sliced along the x -axis so that the area is expressed as a definite integral in terms of x , then the top curve is $y = \ln x$ and the bottom curve is $y = 0$, the x -axis, for $1 \leq x \leq e^2$. Therefore the area would be given by $\int_1^{e^2} (\ln x - 0) \, dx = \int_1^{e^2} \ln x \, dx$. The region could also be sliced along the y -axis so that the area is expressed as a definite integral in terms of y . Then the right curve is the vertical line $x = e^2$ and the left curve is $x = e^y$ for $0 \leq y \leq 2$. In this case, the area would be given by $\int_0^2 (e^2 - e^y) \, dy$. The integral $\int_0^2 e^y \, dy$ would give the area of the region bounded by the graph of $y = \ln x$, the y -axis, and the horizontal line $y = 2$. This would be $2e^2$ minus the area of R .

8-5-7

66. What is the area of the region in the first quadrant bounded on the left by the graph of $x = y^2$ and on the right by the graph of $x = 4y - 3$ for $1 \leq y \leq 3$?

- (A) $\frac{4}{3}$
 (B) $\frac{56}{3}$
 (C) 54
 (D) $\frac{320}{3}$



Answer A

Correct. Since the region is bounded on the left by $x = y^2$ and on the right by $x = 4y - 3$ for $1 \leq y \leq 3$, the area is given by the definite integral $\int_1^3 (4y - 3 - y^2) \, dy$. This is evaluated as follows.

$$\int_1^3 (4y - 3 - y^2) \, dy = \left(2y^2 - 3y - \frac{y^3}{3} \right) \Big|_1^3 = (18 - 9 - 9) - \left(2 - 3 - \frac{1}{3} \right) = \frac{4}{3}$$

67. Let R be the region in the first quadrant bounded above by the graph of $y = \frac{7}{3}x + 1$ and bounded below by the graph of $y = 2^x$ for $0 \leq x \leq 3$. Which of the following definite integrals gives the area of region R ?

- I. $\int_0^3 \left(\left(\frac{7}{3}x + 1 \right) - 2^x \right) \, dx$
 II. $\int_0^3 \left(\frac{\ln y}{\ln 2} - \frac{3}{7}(y - 1) \right) \, dy$
 III. $\int_1^8 \left(\frac{\ln y}{\ln 2} - \frac{3}{7}(y - 1) \right) \, dy$

- (A) None
 (B) I only
 (C) I and II only
 (D) I and III only



Answer D

Correct. The region R is bounded above by the graph of $y = \frac{7}{3}x + 1$ and bounded below by the graph of $y = 2^x$ for $0 \leq x \leq 3$. Therefore, the area of the region is given by the definite integral $\int_0^3 \left(\left(\frac{7}{3}x + 1 \right) - 2^x \right) \, dx$.

Expressed as functions of y , the boundary curves are $x = \frac{\ln y}{\ln 2}$ and $x = \frac{3}{7}(y - 1)$. Region R is bounded to the right by $x = \frac{\ln y}{\ln 2}$ and bounded to the left by $x = \frac{3}{7}(y - 1)$.

. Therefore, the area of the region is given by the definite integral $\int_1^8 \left(\frac{\ln y}{\ln 2} - \frac{3}{7}(y - 1) \right) \, dy$.

8-5-7

68. Let R be the region in the first quadrant bounded above by the graph of $y = 1 + 4x$ and below by the graph of $y = 3^x$ for $0 \leq x \leq 2$. Which of the following definite integrals gives the area of region R ?

I. $\int_0^2 (1 + 4x - 3^x) \, dx$

II. $\int_0^2 \left(\frac{\ln y}{\ln 3} - \frac{y-1}{4} \right) \, dy$

III. $\int_1^9 \left(\frac{\ln y}{\ln 3} - \frac{y-1}{4} \right) \, dy$

- (A) None
- (B) I only
- (C) I and II only
- (D) I and III only

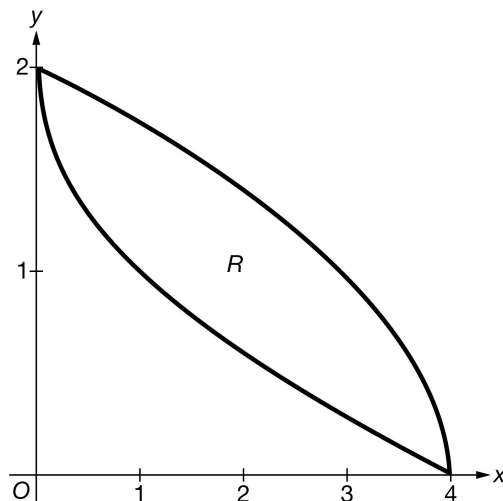
**Answer D**

Correct. The region R is bounded above by the graph of $y = 3^x$ and below by the graph of $y = 1 + 4x$ for $0 \leq x \leq 2$. Therefore, the area of the region is given by the definite integral $\int_0^2 (1 + 4x - 3^x) \, dx$.

Expressed as functions of y , the boundary curves are $x = \frac{y-1}{4}$ and $x = \frac{\ln y}{\ln 3}$. Region R is bounded to the left by $x = \frac{y-1}{4}$ and to the right by $x = \frac{\ln y}{\ln 3}$, for $1 \leq y \leq 9$. Therefore, the area of the region is given by the definite integral $\int_1^9 \left(\frac{\ln y}{\ln 3} - \frac{y-1}{4} \right) \, dy$.

8-5-7

69.



Let R be the region in the first quadrant bounded above by the graph of $y = \frac{4}{\pi} \arccos\left(\frac{x}{4}\right)$ and below by the graph of $y = 2 - \sqrt{x}$, as shown in the figure above. What is the area of the region?

- (A) $\frac{4}{3}$
 (B) $\frac{16}{\pi} + \frac{8}{3}$
 (C) $\frac{16}{\pi} - \frac{8}{3}$ ✓
 (D) $\frac{16}{3}$

Answer C

Correct. Expressed as functions of y , the boundary curves are $x = 4 \cos\left(\frac{\pi y}{4}\right)$ and $x = (y - 2)^2$. Region R is bounded on the right by the graph of $x = 4 \cos\left(\frac{\pi y}{4}\right)$ and bounded on the left by the graph of $x = (y - 2)^2$ for $0 \leq y \leq 2$. Therefore, the area of region R is given by the definite integral $\int_0^2 \left(4 \cos\left(\frac{\pi y}{4}\right) - (y - 2)^2\right) dy$. This is evaluated as follows.

$$\begin{aligned} \int_0^2 \left(4 \cos\left(\frac{\pi y}{4}\right) - (y - 2)^2\right) dy &= \frac{16}{\pi} \sin\left(\frac{\pi y}{4}\right) - \frac{1}{3}(y - 2)^3 \bigg|_{y=0}^{y=2} \\ &= \left(\frac{16}{\pi} \sin\left(\frac{\pi}{2}\right) - 0\right) - \left(\frac{16}{\pi} \sin(0) + \frac{8}{3}\right) = \frac{16}{\pi} - \frac{8}{3} \end{aligned}$$

8-5-7

70. What is the area of the region in the first quadrant bounded on the left by the graph of $x = \frac{y^2}{2}$ and on the right by the graph of $x = 3y - 4$ for $2 \leq y \leq 4$?

- (A) $10 - \frac{56}{6}$ ✓
(B) $10 + \frac{56}{6}$
(C) $\frac{720}{8} - \frac{112}{3}$
(D) $\frac{504}{6} - \frac{192}{2} + 30$

Answer A

Correct. Since the region is bounded on the left by $x = \frac{y^2}{2}$ and on the right by $x = 3y - 4$ for $2 \leq y \leq 4$, the area is given by the definite integral $\int_2^4 \left(3y - 4 - \frac{y^2}{2}\right) dy$. This is evaluated as follows.

$$\int_2^4 \left(3y - 4 - \frac{y^2}{2}\right) dy = \left.\frac{3y^2}{2} - 4y - \frac{y^3}{6}\right|_2^4 = \left(24 - 16 - \frac{64}{6}\right) - \left(6 - 8 - \frac{8}{6}\right) = 10 - \frac{56}{6}$$

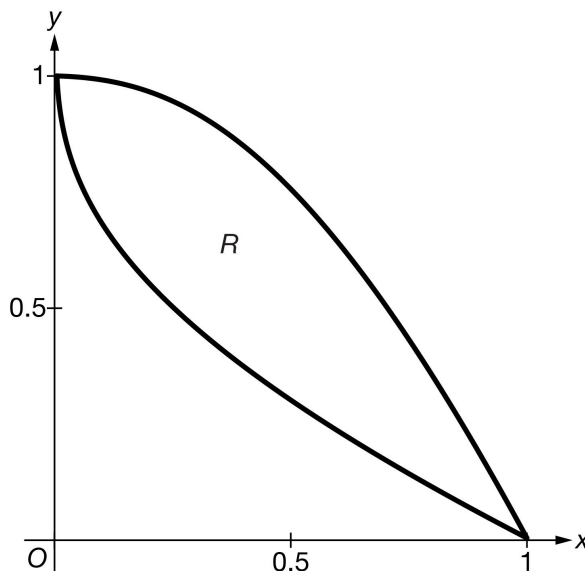
71. Let R be the region bounded by the graph of $x = e^y$, the vertical line $x = 10$, and the horizontal lines $y = 1$ and $y = 2$. Which of the following gives the area of R ?

- (A) $\int_1^2 e^y dy$
(B) $\int_e^{e^2} \ln x dx$
(C) $\int_1^2 (10 - e^y) dy$ ✓
(D) $\int_e^{10} (\ln x - 1) dx$

Answer C

8-5-7

72.



Let R be the region in the first quadrant bounded above by the graph of $y = 1 - x^2$ and below by the graph of $y = 1 - \sqrt{x}$, as shown in the figure above. What is the area of the region?

(A) -1 (B) $\frac{1}{6}$ (C) $\frac{1}{3}$ (D) 1 **Answer C**

Correct. Expressed as functions of y , the boundary curves are $x = \sqrt{1 - y}$ and $x = (1 - y)^2$. Region R is bounded on the right by the graph of $x = \sqrt{1 - y}$ and bounded on the left by the graph of $x = (1 - y)^2$ for $0 \leq y \leq 1$. Therefore, the area of region R is given by the definite integral $\int_0^1 \left(\left(\sqrt{1 - y} \right) - (1 - y)^2 \right) dy$. This is evaluated as follows.

$$\int_0^1 \left(\left(\sqrt{1 - y} \right) - (1 - y)^2 \right) dy = (-1) \frac{2}{3} (1 - y)^{\frac{3}{2}} - (-1) \frac{(1 - y)^3}{3} \Big|_0^1 = 0 - \left(-\frac{2}{3} + \frac{1}{3} \right) = \frac{1}{3}$$

8-5-7

73. The base of a solid is the region in the first quadrant bounded by the x -axis, the y -axis, and the graph of $y = 4 - x^2$. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 6 times the x -coordinate. What is the volume of the solid?
- (A) 16
- (B) 24
- (C) 32
- (D) 64

Answer B

Correct. The cross section perpendicular to the x -axis at a point with x -coordinate x is a rectangle with width $y = 4 - x^2$ and height $6x$. The area of the rectangle is therefore $y \cdot 6x = 6x(4 - x^2) = 24x - 6x^3$. The volume of the solid is found using the definite integral of the cross-sectional area. The graph of $y = 4 - x^2$ intersects the x -axis in the first quadrant at $x = 2$ so cross sections can be taken from $x = 0$ to $x = 2$.

$$\int_0^2 (24x - 6x^3) dx = 12x^2 - \frac{3}{2}x^4 \Big|_0^2 = 48 - 24 = 24$$

74. Let R be the region in the first quadrant bounded by the graph of $y = e^{-x}$, the x -axis, the y -axis, and the line $x = b$. Region R is the base of a solid whose cross sections perpendicular to the x -axis are squares. If the volume of the solid is $\frac{4}{9}$, what is the value of b ?
- (A) $\ln\left(\frac{3}{\sqrt{17}}\right)$
- (B) $\ln\left(\frac{3}{\sqrt{7}}\right)$
- (C) $\ln\left(\frac{9}{5}\right)$
- (D) $\frac{1}{2}\ln 9$

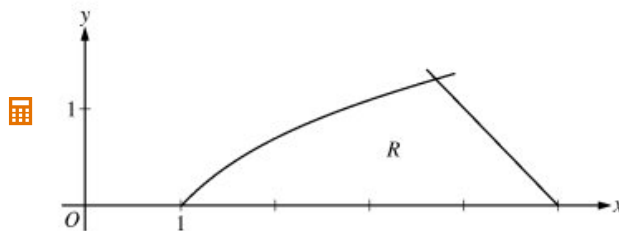
Answer D

Correct. The area of a square of side length s is s^2 . A typical cross section of the solid is a square with one side from the x -axis to the graph of $y = e^{-x}$. The length of the side of the square is therefore $s = e^{-x}$, so the area of the square is $(e^{-x})^2 = e^{-2x}$. The volume of the solid is found using the definite integral of the cross-sectional area.

$$\begin{aligned} \frac{4}{9} &= \int_0^b e^{-2x} dx = -\frac{1}{2}e^{-2x} \Big|_0^b = -\frac{1}{2}e^{-2b} + \frac{1}{2} \\ \frac{1}{2}e^{-2b} &= \frac{1}{2} - \frac{4}{9} = \frac{1}{18} \Rightarrow e^{-2b} = \frac{1}{9} \Rightarrow b = -\frac{1}{2}\ln\left(\frac{1}{9}\right) = \frac{1}{2}\ln 9 \end{aligned}$$

8-5-7

75.



Let R be the region in the first quadrant bounded by the x -axis and the graphs of $y = \ln x$ and $y = 5 - x$, as shown in the figure above.

(a) Find the area of R .

(b) Region R is the base of a solid. For the solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

(c) The horizontal line $y = k$ divides R into two regions of equal area. Write, but do not solve, an equation involving one or more integrals whose solution gives the value of k .

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\begin{aligned} \text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

OR

$$\begin{aligned} \text{Area} &= \int_1^A \ln x \, dx + \int_A^5 (5 - x) dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

✓

0	1	2	3
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The student response earns three of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

8-5-7

$$\begin{aligned}\text{Area} &= \int_0^B (5 - y - e^y) dy \\ &= 2.986 \text{ (or 2.985)}\end{aligned}$$

OR

$$\begin{aligned}\text{Area} &= \int_1^A \ln x \, dx + \int_A^5 (5 - x) dx \\ &= 2.986 \text{ (or 2.985)}\end{aligned}$$

Part B

2 points are earned for the integrands

1 point is earned for the expression for total volume

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\text{Volume} = \int_1^A (\ln x)^2 dx + \int_A^5 (5 - x)^2 dx$$



0	1	2	3
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The student response earns three of the following points:

2 points are earned for the integrands

1 point is earned for the expression for total volume

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\text{Volume} = \int_1^A (\ln x)^2 dx + \int_A^5 (5 - x)^2 dx$$

Part C

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the equation

$$\ln x = 5 - x \Rightarrow x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

8-5-7

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \left(\text{or } \frac{1}{2} \cdot 2.985 \right)$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the equation

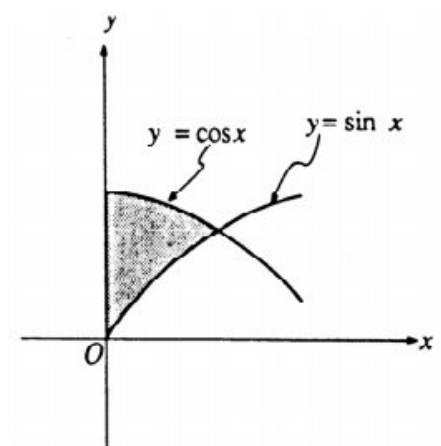
$$\ln x = 5 - x \approx x = 3.69344$$

Therefore, the graphs of $y = \ln x$ and $y = 5 - x$ intersect in the first quadrant at the point $(A, B) = (3.69344, 1.30656)$.

$$\int_0^k (5 - y - e^y) dy = \frac{1}{2} \cdot 2.986 \left(\text{or } \frac{1}{2} \cdot 2.985 \right)$$

8-5-7

76.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = \sin x$ and $y = \cos x$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Find the volume of the solid whose base is R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi/4} \cos x - \sin x \, dx \\
 &= (\sin x + \cos x) \Big|_0^{\pi/4} \\
 &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\
 &= \sqrt{2} - 1
 \end{aligned}$$

✓

0	1	2	3
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The student response earns all of the following points:

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

8-5-7

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi/4} \cos x - \sin x \, dx \\
 &= (\sin x + \cos x) \Big|_0^{\pi/4} \\
 &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\
 &= \sqrt{2} - 1
 \end{aligned}$$

Part B

1 point is earned for limits

1 point is earned for integrand

1 point is **deducted** for constant error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for integrand

1 point is **deducted** for constant error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$

Part C

2 points are earned for integral

8-5-7

1 point is **deducted** for constant or limit error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \int_0^{\pi/4} (\cos x - \sin x)^2 dx \\
 &= \int_0^{\pi/4} 1 - 2 \sin x \cos x dx \\
 &= (x - \sin^2 x) \Big|_0^{\pi/4} \\
 &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\
 &= \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$



0	1	2	3
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The student response earns all of the following points:

2 points are earned for integral

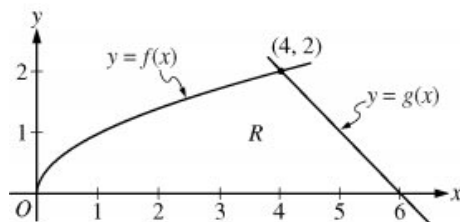
1 point is **deducted** for constant or limit error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \int_0^{\pi/4} (\cos x - \sin x)^2 dx \\
 &= \int_0^{\pi/4} 1 - 2 \sin x \cos x dx \\
 &= (x - \sin^2 x) \Big|_0^{\pi/4} \\
 &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\
 &= \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$

8-5-7

77.



The functions f and g are given by $f(x) = \sqrt{x}$ and $g(x) = 6 - x$. Let R be the region bounded by the x -axis and the graphs of f and g , as shown in the figure above.

(a) Find the area of R .

(b) The region R is the base of a solid. For each y , where $0 \leq y \leq 2$, the cross section of the solid taken perpendicular to the y -axis is a rectangle whose base lies in R and whose height is $2y$. Write, but do not evaluate, an integral expression that gives the volume of the solid.

(c) There is a point P on the graph of f at which the line tangent to the graph of f is perpendicular to the graph of g . Find the coordinates of point P .

Part A

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^4 \sqrt{x} \, dx + \frac{1}{2} \cdot 2 \cdot 2 = \frac{2}{3} x^{3/2} \Big|_{x=0}^{x=4} + 2 = \frac{22}{3}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^4 \sqrt{x} \, dx + \frac{1}{2} \cdot 2 \cdot 2 = \frac{2}{3} x^{3/2} \Big|_{x=0}^{x=4} + 2 = \frac{22}{3}$$

Part B

2 points are earned for integrand

1 point is earned for the answer

8-5-7

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

$$y = \sqrt{x} \Rightarrow x = y^2$$

$$y = 6 - x \Rightarrow x = 6 - y$$

$$\text{Width} = (6 - y) - y^2$$

$$\text{Volume} = \int_0^2 2y(6 - y - y^2) dy$$

Part C

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

The point P has coordinates $(\frac{1}{4}, \frac{1}{2})$.

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for the equation

1 point is earned for the answer

8-5-7

$$g'(x) = -1$$

Thus a line perpendicular to the graph of g has slope 1.

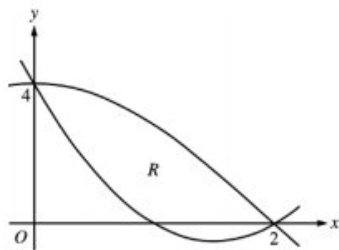
$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\frac{1}{2\sqrt{x}} = 1 \Rightarrow x = \frac{1}{4}$$

The point P has coordinates $\left(\frac{1}{4}, \frac{1}{2}\right)$.

8-5-7

78.



Let $f(x) = 2x^2 - 6x + 4$ and $g(x) = 4\cos(1/4 \pi x)$. Let R be the region bounded by the graphs of f and g , as shown in the figure above.

(a) Find the area of R .

(b) Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 4$.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for the integrand

2 points are earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^2 [g(x) - f(x)] dx \\
 &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx \\
 &= \left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2 \\
 &= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3}
 \end{aligned}$$

✓

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned for the antiderivative

1 point is earned for the answer

8-5-7

$$\begin{aligned}
 \text{Area} &= \int_0^2 [g(x) - f(x)] dx \\
 &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx \\
 &= \left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2 \\
 &= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3}
 \end{aligned}$$

Part B

2 points are earned for the integrand.

1 point is earned for the limits and constant.

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^2 \left[(4 - f(x))^2 - (4 - g(x))^2 \right] dx \\
 &= \pi \int_0^2 \left[(4 - (2x^2 - 6x + 4))^2 - \left(4 - 4 \cos\left(\frac{\pi}{4}x\right)\right)^2 \right] dx
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand.

1 point is earned for the limits and constant.

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^2 \left[(4 - f(x))^2 - (4 - g(x))^2 \right] dx \\
 &= \pi \int_0^2 \left[(4 - (2x^2 - 6x + 4))^2 - \left(4 - 4 \cos\left(\frac{\pi}{4}x\right)\right)^2 \right] dx
 \end{aligned}$$

Part C

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\begin{aligned}
 \text{Volume} &= \int_0^2 [g(x) - f(x)]^2 dx \\
 &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right]^2 dx
 \end{aligned}$$

8-5-7



0	1	2
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The student response earns all of the following points:

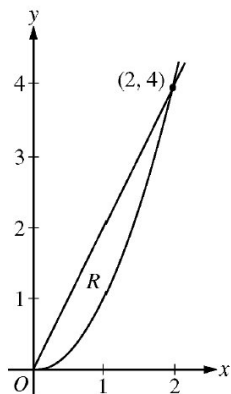
1 point is earned for the integrand

1 point is earned for the limits and constant

$$\begin{aligned}\text{Volume} &= \int_0^2 [g(x) - f(x)] \, dx \\ &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] \, dx\end{aligned}$$

8-5-7

79.



Let R be the region in the first quadrant enclosed by the graphs of $y = 2x$ and $y = x^2$, as shown in the figure above.

(a) Find the area of R .

(b) The region R is the base of a solid. For this solid, at each x the cross section perpendicular to the x -axis has area $A(x) = \sin(\pi/2x)$. Find the volume of the solid.

(c) Another solid has the same base R . For this solid, the cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an integral expression for the volume of the solid.

Part A

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^2 (2x - x^2) dx \\ &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\ &= \frac{4}{3}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^2 (2x - x^2) dx \\ &= x^2 - \frac{1}{3}x^3 \Big|_{x=0}^{x=2} \\ &= \frac{4}{3}\end{aligned}$$

8-5-7

Part B

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi}\end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^2 \sin\left(\frac{\pi}{2}x\right) dx \\ &= -\frac{2}{\pi} \cos\left(\frac{\pi}{2}x\right) \Big|_{x=0}^{x=2} \\ &= \frac{4}{\pi}\end{aligned}$$

Part C

2 points are earned for the integrand

1 point is earned for the limit

$$\text{Volume} = \int_0^4 \left(\sqrt{y} - \frac{y}{2}\right)^2 dy$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

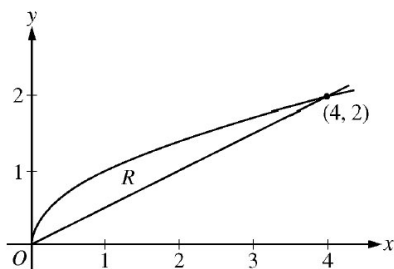
1 point is earned for the limit

8-5-7

$$\text{Volume} = \int_0^4 \left(\sqrt{y} - \frac{y}{2} \right)^2 dy$$

8-5-7

80.



Let R be the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$, as shown in the figure above.

(a) Find the area of R .

(b) The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are squares. Find the volume of this solid.

(c) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 2$.

Part A

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{3/2} - \frac{x^2}{4} \Big|_{x=0}^{x=4} = \frac{4}{3}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{3/2} - \frac{x^2}{4} \Big|_{x=0}^{x=4} = \frac{4}{3}$$

Part B

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\begin{aligned} \text{Volume} &= \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right)^2 dx = \int_0^4 \left(x - x^{3/2} + \frac{x^2}{4} \right) dx \\ &= \frac{x^2}{2} - \frac{2x^{5/2}}{5} + \frac{x^3}{12} \Big|_{x=0}^{x=4} = \frac{8}{15} \end{aligned}$$

8-5-7

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\begin{aligned}\text{Volume} &= \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right)^2 dx = \int_0^4 \left(x - x^{3/2} + \frac{x^2}{4} \right) dx \\ &= \frac{x^2}{2} - \frac{2x^{5/2}}{5} + \frac{x^3}{12} \Big|_{x=0}^{x=4} = \frac{8}{15}\end{aligned}$$

Part C

1 point is earned for the limits and constant

2 points are earned for the integrand

$$\text{Volume} = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

2 points are earned for the integrand

$$\text{Volume} = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$

8-5-7

81.  Let R be the region in the first quadrant bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the vertical line $x = -1$.
- (c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the y -axis are squares. Find the volume of this solid.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0, 0)$ and $(9, 3)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0, 0)$ and $(9, 3)$.

Part B

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

8-5-7

$$\pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy$$

$$= \frac{207\pi}{5} = 130.061 \text{ or } 130.062$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

$$\pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy$$

$$= \frac{207\pi}{5} = 130.061 \text{ or } 130.062$$

Part C

1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

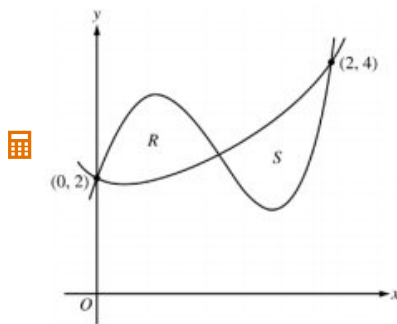
1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$

8-5-7

82.



Let f and g be the functions defined by $f(x) = 1 + x + e^{x^2-2x}$ and $g(x) = x^4 - 6.5x^2 + 6x + 2$. Let R and S be the two regions enclosed by the graphs of f and g shown in the figure above.

(a) Find the sum of the areas of regions R and S .

(b) Region S is the base of a solid whose cross sections perpendicular to the x -axis are squares. Find the volume of the solid.

(c) Let h be the vertical distance between the graphs of f and g in region S . Find the rate at which h changes with respect to x when $x = 1.8$.

Part A

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y = f(x)$ and $y = g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and

$(A, B) = (1.032832, 2.401108)$.

$$\begin{aligned} \text{Area} &= \int_0^A [g(x) - f(x)]dx + \int_A^2 [f(x) - g(x)]dx \\ &= 0.997427 + 1.006919 = 2.004 \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y = f(x)$ and $y = g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and

$(A, B) = (1.032832, 2.401108)$.

$$\begin{aligned} \text{Area} &= \int_0^A [g(x) - f(x)]dx + \int_A^2 [f(x) - g(x)]dx \\ &= 0.997427 + 1.006919 = 2.004 \end{aligned}$$

Part B

8-5-7

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$

Part C

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$

8-5-7

83.  Let R be the region in the first quadrant under the graph of $y = \frac{1}{\sqrt{x}}$ for $4 \leq x \leq 9$.

(a) Find the area of R .

(b) If the line $x = k$ divides the region R into two regions of equal area, what is the value of k ?

(c) Find the volume of the solid whose base is the region R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part A

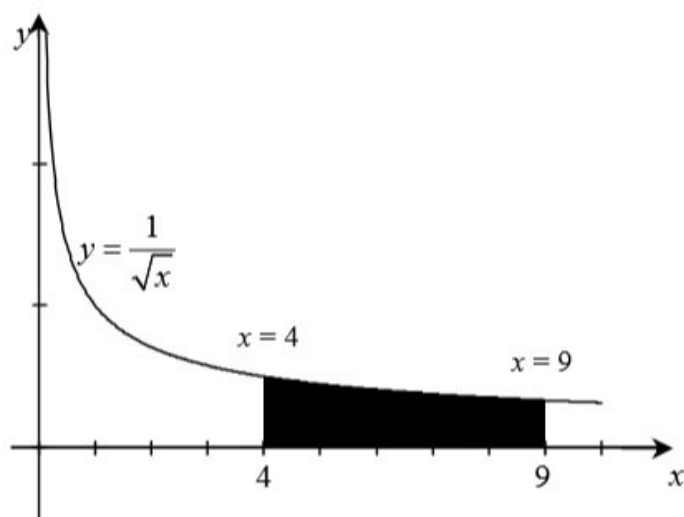
1 point is earned for the correct limits in x

1 point is earned for the correct integrand including constant must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$

0/1 if the integrand is incorrect, wrong limits are read



✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits in x

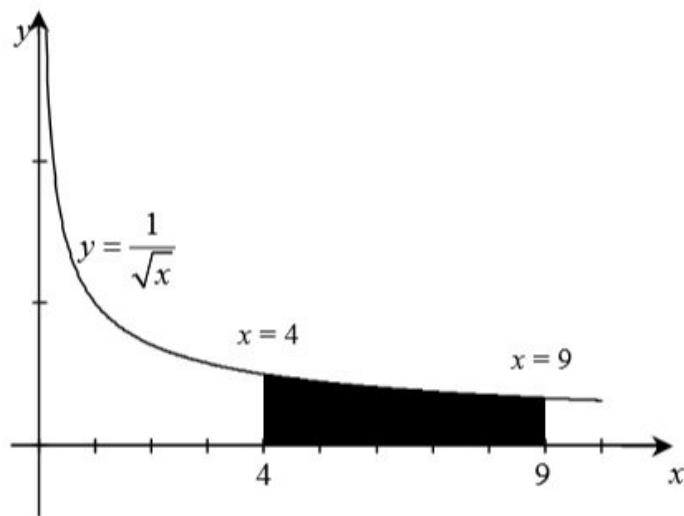
1 point is earned for the correct integrand including constant must be $\frac{1}{\sqrt{x}}$

1 point is earned for the correct answer

$$\int_4^9 \frac{dx}{\sqrt{x}} = 2$$

0/1 if the integrand is incorrect, wrong limits are read

8-5-7

**Part B**

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x} \Big|_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct demonstration of variable limit using k as upper or lower limit with 4 or 9 (or imported limit)

$$\int_4^k \frac{dx}{\sqrt{x}} \text{ or } \int_k^9 \frac{dx}{\sqrt{x}}$$

1 point is earned for the correct equation involving the two halves of R that when solved will give answer with demonstrated algorithm that knows when to stop

8-5-7

1 point is earned for the correct answer

0/1 if answer from equation not involving relevant areas

1/3 bald answer

$$\int_4^k \frac{dx}{\sqrt{x}} = 1$$

$$2\sqrt{x}]_4^k = 1$$

$$2\sqrt{k} - 2\sqrt{4} = 1$$

$$k = \frac{25}{4}$$

$$\left(\text{or } \int_k^9 \frac{dx}{\sqrt{x}} = 1 \text{ or } \int_4^k \frac{dx}{\sqrt{x}} = \int_k^9 \frac{dx}{\sqrt{x}} \right)$$

Part C

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

1 point is earned for the correct answer

0/1 if integrand is incorrect

$$\begin{aligned} \text{Volume} &= \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx \\ &= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4} \end{aligned}$$

(or 0.811)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct indefinite limits using x

1 point is earned for the correct integrand including constant

1 point is earned for the correct answer

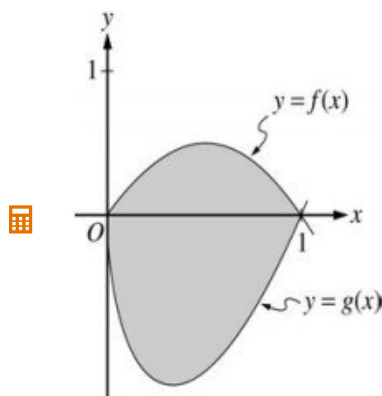
0/1 if integrand is incorrect

$$\begin{aligned} \text{Volume} &= \int_4^9 \left(\frac{1}{\sqrt{x}} \right)^2 dx \\ &= \int_4^9 \frac{dx}{x} = \ln x \Big|_4^9 = \ln \frac{9}{4} \end{aligned}$$

(or 0.811)

8-5-7

84.



Let f and g be the functions given by $f(x) = 2x(1 - x)$ and $g(x) = 3(x - 1)\sqrt{x}$ for $0 \leq x \leq 1$. The graphs of f and g are shown in the figure above.

- (a) Find the area of the shaded region enclosed by the graphs of f and g .
- (b) Find the volume of the solid generated when the shaded region enclosed by the graphs of f and g is revolved about the horizontal line $y = 2$.
- (c) Let h be the function given by $h(x) = kx(1 - x)$ for $0 \leq x \leq 1$. For each $k > 0$, the region (not shown) enclosed by the graphs of h and g is the base of a solid with square cross sections perpendicular to the x -axis. There is a value of k for which the volume of this solid is equal to 15. Write, but do not solve, an equation involving an integral expression that could be used to find the value of k .

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1 - x) - 3(x - 1)\sqrt{x}) dx = 1.133 \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1 - x) - 3(x - 1)\sqrt{x}) dx = 1.133 \end{aligned}$$

Part B

8-5-7

1 point is earned for the limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\ &= \pi \int_0^1 \left((2 - 3(x - 1)\sqrt{x})^2 - (2 - 2x(1 - x))^2 \right) dx \\ &= 16.179\end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\ &= \pi \int_0^1 \left((2 - 3(x - 1)\sqrt{x})^2 - (2 - 2x(1 - x))^2 \right) dx \\ &= 16.179\end{aligned}$$

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^1 (h(x) - g(x))^2 dx \\ &= \int_0^1 (kx(1 - x) - 3(x - 1)\sqrt{x})^2 dx = 15\end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

8-5-7

2 points are earned for the integrand

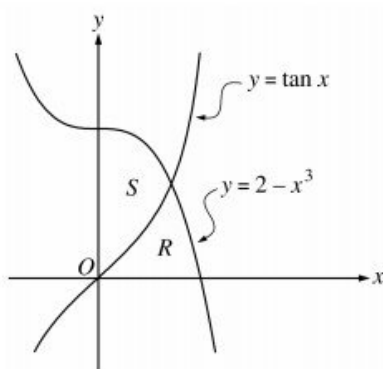
1 point is earned for the answer

$$\text{Volume} = \int_0^1 (h(x) - g(x))^2 dx$$

$$\int_0^1 (kx(1-x) - 3(x-1)\sqrt{x})^2 dx = 15$$

8-5-7

85.



Let R and S be the regions in the first quadrant shown in the figure above. The region R is bounded by the x -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$. The region S is bounded by the y -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$.

- Find the area of R .
- Find the area of S .
- Find the volume of the solid generated when S is revolved about the x -axis.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

$$2 - x^3 = \tan x \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area } R = \int_0^A \tan x \, dx + \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx = 0.729$$

or

$$\text{Area } R = \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy = 0.729$$

or

$$\text{Area } R = \int_0^{\sqrt[3]{2}} (2 - x^3) \, dx - \int_0^A (2 - x^3 - \tan x) \, dx = 0.729$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

8-5-7

$$2 - x^3 = \tan x \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area } R = \int_0^A \tan x \, dx + \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx = 0.729$$

or

$$\text{Area } R = \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy = 0.729$$

or

$$\text{Area } R = \int_0^{\sqrt[3]{2}} (2 - x^3) \, dx - \int_0^A (2 - x^3 - \tan x) \, dx = 0.729$$

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area } S = \int_0^A (2 - x^3 - \tan x) \, dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area } S = \int_0^B \tan^{-1} y \, dy + \int_B^2 (2 - y)^{1/3} \, dy = 1.160 \text{ or } 1.161$$

or

Area S

$$= \int_0^2 (2 - y)^{1/3} \, dy - \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy$$

$$= 1.160 \text{ or } 1.161$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

8-5-7

$$\text{Area } S = \int_0^A (2 - x^3 - \tan x) dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area } S = \int_0^B \tan^{-1} y \, dy + \int_B^2 (2 - y)^{1/3} dy = 1.160 \text{ or } 1.161$$

or

Area S

$$= \int_0^2 (2 - y)^{1/3} dy - \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy$$

$$= 1.160 \text{ or } 1.161$$

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx$$

$$= 2.652\pi \text{ or } 8.331 \text{ or } 8.332$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

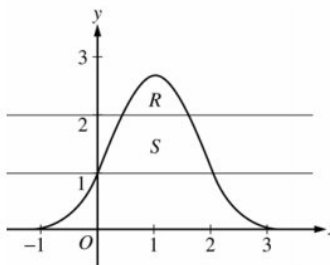
1 point is earned for the answer

$$\text{Volume} = \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx$$

$$= 2.652\pi \text{ or } 8.331 \text{ or } 8.332$$

8-5-7

86.



Let R be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal line $y = 2$, and let S be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal lines $y = 1$ and $y = 2$, as shown above.

(a) Find the area of R .

(b) Find the area of S .

(c) Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2)dx = 0.514$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2)dx = 0.514$$

8-5-7

Part B

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$e^{2x-x^2} = 1 \text{ when } x = 0, 2$$

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1)dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

OR

$$\int_0^P (e^{2x-x^2} - 1)dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1)dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$e^{2x-x^2} = 1 \text{ when } x = 0, 2$$

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1)dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

8-5-7

OR

$$\int_0^P (e^{2x-x^2} - 1)dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1)dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$

Part C

2 points are earned for the integrand

1 point is earned for the constant and limits

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Volume} = \pi \int_P^Q \left((e^{2x-x^2} - 1)^2 - (2 - 1)^2 \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the constant and limits

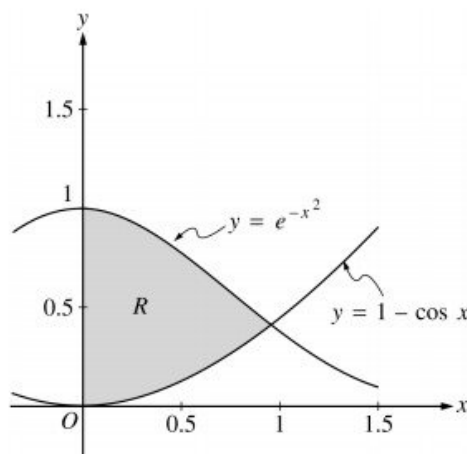
$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Volume} = \pi \int_P^Q \left((e^{2x-x^2} - 1)^2 - (2 - 1)^2 \right) dx$$

8-5-7

87.



Let R be the shaded region in the first quadrant enclosed by the graphs of $y = e^{-x^2}$, $y = 1 - \cos x$, and the y -axis, as shown in the figure above.

(a) Find the area of the region R .

(b) Find the volume of the solid generated when the region R is revolved about the x -axis.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.

Part A

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$e^{-x^2} = 1 - \cos x \text{ at } x = 0.941944 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A (e^{-x^2} - (1 - \cos x)) dx \\ &= 0.590 \text{ or } 0.591 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$e^{-x^2} = 1 - \cos x \text{ at } x = 0.941944 = A$$

8-5-7

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^A \left(e^{-x^2} - (1 - \cos x) \right) dx \\ &= 0.590 \text{ or } 0.591\end{aligned}$$

Part B

2 points are earned for the integrand and constant

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left(\left(e^{-x^2} \right)^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746 \text{ or } 1.747\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand and constant

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left(\left(e^{-x^2} \right)^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746 \text{ or } 1.747\end{aligned}$$

Part C

2 points are earned for the integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^A \left(e^{-x^2} - (1 - \cos x) \right)^2 dx \\ &= 0.461\end{aligned}$$

8-5-7



0	1	2	3
---	---	---	---

The student response earns all of the following points:

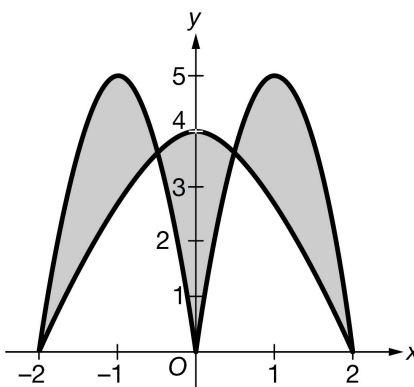
2 points are earned for the integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^A \left(e^{-x^2} - (1 - \cos x) \right)^2 dx \\ &= 0.461\end{aligned}$$

88.



The figure above shows the graphs of $y = |5 \sin(\frac{\pi}{2}x)|$ and $y = 4 - x^2$ for $-2 \leq x \leq 2$. The x -coordinates of the points of intersection of the graphs between -2 and 2 are x_1 and x_2 , where $-2 < x_1 < x_2 < 2$. Which of the following gives the sum of the areas of the shaded regions?

(A) $\int_{-2}^{x_1} (4 - x^2) \, dx + \int_{x_1}^{x_2} |5 \sin(\frac{\pi}{2}x)| \, dx + \int_{x_2}^2 (4 - x^2) \, dx$

(B) $\int_{-2}^2 \left((4 - x^2) - |5 \sin(\frac{\pi}{2}x)| \right) \, dx$

(C) $\int_{-2}^{x_1} |5 \sin(\frac{\pi}{2}x)| \, dx + \int_{x_1}^{x_2} (4 - x^2) \, dx + \int_{x_2}^2 |5 \sin(\frac{\pi}{2}x)| \, dx$

(D) $\int_{-2}^{x_1} \left(|5 \sin(\frac{\pi}{2}x)| - (4 - x^2) \right) \, dx + \int_{x_1}^{x_2} \left((4 - x^2) - |5 \sin(\frac{\pi}{2}x)| \right) \, dx + \int_{x_2}^2 \left(|5 \sin(\frac{\pi}{2}x)| - (4 - x^2) \right) \, dx$ ✓

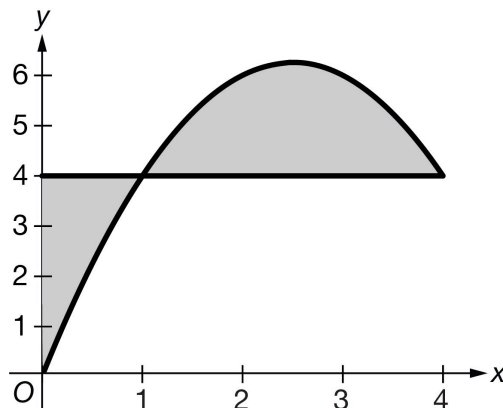
Answer D

Correct. For $-2 \leq x \leq x_1$ and $x_2 \leq x \leq 2$, the regions are bounded above by the graph of $y = |5 \sin(\frac{\pi}{2}x)|$ and bounded below by the graph of $y = 4 - x^2$. For $x_1 \leq x \leq x_2$, the region is bounded above by the graph of $y = 4 - x^2$ and bounded below by the graph of $y = |5 \sin(\frac{\pi}{2}x)|$. Therefore, the sum of the areas is given by

$$\int_{-2}^{x_1} \left(|5 \sin(\frac{\pi}{2}x)| - (4 - x^2) \right) \, dx + \int_{x_1}^{x_2} \left((4 - x^2) - |5 \sin(\frac{\pi}{2}x)| \right) \, dx + \int_{x_2}^2 \left(|5 \sin(\frac{\pi}{2}x)| - (4 - x^2) \right) \, dx.$$

8-5-7

89.



The figure above shows the graph of $y = 5x - x^2$ and the line $y = 4$ for $0 \leq x \leq 4$. What is the sum of the areas of the shaded regions?

(A) $\frac{8}{3}$

(B) $\frac{9}{2}$

(C) $\frac{19}{3}$

(D) $\frac{104}{3}$

**Answer C**

Correct. For $0 \leq x \leq 1$, the region is bounded above by the graph of $y = 4$ and below by the graph of $y = 5x - x^2$. For $1 \leq x \leq 4$, the region is bounded above by the graph of $y = 5x - x^2$ and below by the graph of $y = 4$. Therefore the sum of the areas is given by

$\int_0^1 (4 - (5x - x^2)) \, dx + \int_1^4 ((5x - x^2) - 4) \, dx$. This is evaluated as follows.

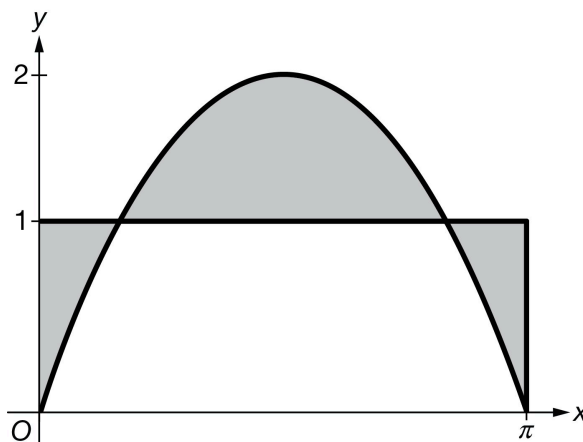
$$\int_0^1 (4 - (5x - x^2)) \, dx = \left(4x - \frac{5}{2}x^2 + \frac{1}{3}x^3 \right) \Big|_0^1 = \left(4 - \frac{5}{2} + \frac{1}{3} \right) - 0 = \frac{11}{6}$$

$$\int_1^4 ((5x - x^2) - 4) \, dx = \left(\frac{5}{2}x^2 - \frac{1}{3}x^3 - 4x \right) \Big|_1^4 = \left(40 - \frac{64}{3} - 16 \right) - \left(\frac{5}{2} - \frac{1}{3} - 4 \right) = \frac{9}{2}$$

$$\int_0^1 (4 - (5x - x^2)) \, dx + \int_1^4 ((5x - x^2) - 4) \, dx = \frac{11}{6} + \frac{9}{2} = \frac{19}{3}$$

8-5-7

90.



The figure above shows the graph of $y = 2 \sin x$ and the line $y = 1$ for $0 \leq x \leq \pi$. Which of the following gives the sum of the areas of the shaded regions?

- (A) $\int_0^\pi (2 \sin x - 1) \, dx$
- (B) $\int_0^{\frac{\pi}{6}} 2 \sin x \, dx + \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 1 \, dx + \int_{\frac{5\pi}{6}}^\pi 2 \sin x \, dx$
- (C) $\int_0^{\frac{\pi}{6}} (1 - 2 \sin x) \, dx + \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (2 \sin x - 1) \, dx + \int_{\frac{5\pi}{6}}^\pi (1 - 2 \sin x) \, dx$ ✓
- (D) $\int_0^\pi (2 \sin x + 1) \, dx$

Answer C

Correct. The equation $1 = 2 \sin x$ is solved for $0 \leq x \leq \pi$, in order to determine the x -coordinates of the points of intersection of the graphs.

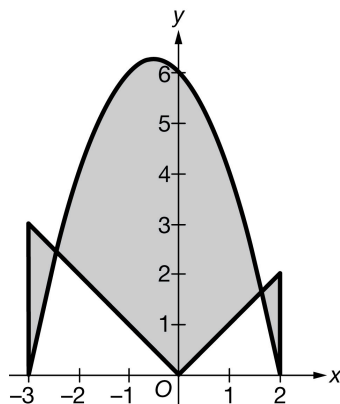
$$1 = 2 \sin x \Rightarrow \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6} \text{ and } x = \frac{5\pi}{6}$$

For $0 \leq x \leq \frac{\pi}{6}$ and $\frac{5\pi}{6} \leq x \leq \pi$, the regions are bounded above by the graph of $y = 1$ and below by the graph of $y = 2 \sin x$. For $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$, the region is bounded above by the graph of $y = 2 \sin x$ and below by the graph of $y = 1$. Therefore the sum of the areas is

$$\text{given by } \int_0^{\frac{\pi}{6}} (1 - 2 \sin x) \, dx + \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (2 \sin x - 1) \, dx + \int_{\frac{5\pi}{6}}^\pi (1 - 2 \sin x) \, dx.$$

8-5-7

91.



The figure above shows the graphs of $y = |x|$ and $y = 6 - x - x^2$ for $-3 \leq x \leq 2$. The x -coordinates of the points of intersection of the graphs are x_1 and x_2 , where $x_1 < x_2$. Which of the following gives the sum of the areas of the shaded regions?

(A) $\int_{-3}^{x_1} (6 - x - x^2) \, dx + \int_{x_1}^{x_2} |x| \, dx + \int_{x_2}^2 (6 - x - x^2) \, dx$

(B) $\int_{-3}^2 ((6 - x - x^2) - |x|) \, dx$

(C) $\int_{-3}^{x_1} |x| \, dx + \int_{x_1}^{x_2} (6 - x - x^2) \, dx + \int_{x_2}^2 |x| \, dx$

(D) $\int_{-3}^{x_1} (|x| - (6 - x - x^2)) \, dx + \int_{x_1}^{x_2} ((6 - x - x^2) - |x|) \, dx + \int_{x_2}^2 (|x| - (6 - x - x^2)) \, dx$ ✓

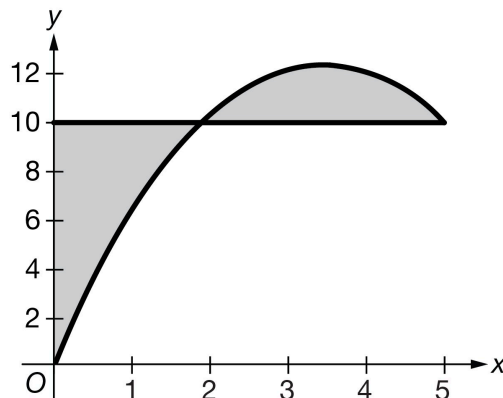
Answer D

Correct. For $-3 \leq x \leq x_1$ and $x_2 \leq x \leq 2$, the regions are bounded above by the graph of $y = |x|$ and below by the graph of $y = 6 - x - x^2$. For $x_1 \leq x \leq x_2$, the region is bounded above by the graph of $y = 6 - x - x^2$ and below by the graph of $y = |x|$. Therefore, the sum of the areas is given by

$$\int_{-3}^{x_1} (|x| - (6 - x - x^2)) \, dx + \int_{x_1}^{x_2} ((6 - x - x^2) - |x|) \, dx + \int_{x_2}^2 (|x| - (6 - x - x^2)) \, dx.$$

8-5-7

92.



The figure above shows the graph of $y = 7x - x^2$ and the line $y = 10$ for $0 \leq x \leq 5$. Which of the following gives the sum of the areas of the shaded regions?

- (A) $50 - \frac{175}{2} + \frac{125}{3}$
 (B) $\frac{175}{2} - \frac{117}{3} - 44$
 (C) $\frac{175}{2} - \frac{109}{3} - 38$ ✓
 (D) $\frac{175}{2} - \frac{125}{3} + 50$

Answer C

Correct. For $0 \leq x \leq 2$, the region is bounded above by the graph of $y = 10$ and below by the graph of $y = 7x - x^2$. For $2 \leq x \leq 5$, the region is bounded above by the graph of $y = 7x - x^2$ and below by the graph of $y = 10$. Therefore, the sum of the areas is given by

$\int_0^2 (10 - (7x - x^2)) \, dx + \int_2^5 ((7x - x^2) - 10) \, dx$. This is evaluated as follows.

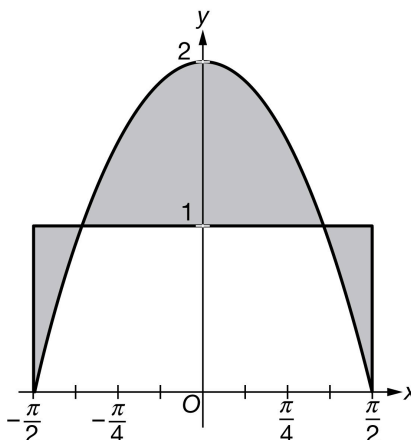
$$\int_0^2 (10 - 7x + x^2) \, dx = 10x - \frac{7x^2}{2} + \frac{x^3}{3} \Big|_0^2 = 20 - 14 + \frac{8}{3} - 0 = 6 + \frac{8}{3}$$

$$\int_2^5 (7x - x^2 - 10) \, dx = \frac{7x^2}{2} - \frac{x^3}{3} - 10x \Big|_2^5 = \left(\frac{7(25)}{2} - \frac{125}{3} - 50 \right) - \left(14 - \frac{8}{3} - 20 \right) = \frac{175}{2} - \frac{117}{3} - 44$$

$$\int_0^2 (10 - 7x + x^2) \, dx + \int_2^5 (7x - x^2 - 10) \, dx = \left(6 + \frac{8}{3} \right) + \left(\frac{175}{2} - \frac{117}{3} - 44 \right) = \frac{175}{2} - \frac{109}{3} - 38$$

8-5-7

93.



The figure above shows the graph of $y = 2 \cos x$ and the line $y = 1$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. Which of the following gives the sum of the areas of the shaded regions?

(A) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2 \cos x - 1) \, dx$

(B) $\int_{-\frac{\pi}{2}}^{-\frac{\pi}{3}} 2 \cos x \, dx + \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} 1 \, dx + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 2 \cos x \, dx$

(C) $\int_{-\frac{\pi}{2}}^{-\frac{\pi}{3}} (1 - 2 \cos x) \, dx + \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} (2 \cos x - 1) \, dx + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (1 - 2 \cos x) \, dx$ ✓

(D) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + 2 \cos x) \, dx$

Answer C

Correct. The equation $1 = 2 \cos x$ is solved for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ in order to determine the x -coordinates of the points of intersection of the graphs.

$$1 = 2 \cos x \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = -\frac{\pi}{3} \text{ or } x = \frac{\pi}{3}$$

For $-\frac{\pi}{2} \leq x \leq -\frac{\pi}{3}$ and $\frac{\pi}{3} \leq x \leq \frac{\pi}{2}$, the regions are bounded above by the graph of $y = 1$ and below by the graph of $y = 2 \cos x$. For $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$, the region is bounded above by the graph of $y = 2 \cos x$ and below by the graph of $y = 1$. Therefore, the sum of the areas is

given by $\int_{-\frac{\pi}{2}}^{-\frac{\pi}{3}} (1 - 2 \cos x) \, dx + \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} (2 \cos x - 1) \, dx + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (1 - 2 \cos x) \, dx$.

8-5-7

94. How many vertical asymptotes does the graph of $y = \frac{x-2}{x^4-16}$ have?

- (A) One
(B) Two
(C) Three
(D) Four

**Answer A**

Correct. Since $\frac{x-2}{x^4-16} = \frac{x-2}{(x^2-4)(x^2+4)} = \frac{x-2}{(x-2)(x+2)(x^2+4)} = \frac{1}{(x+2)(x^2+4)}$ for $x \neq 2$, the graph has only one vertical asymptote, which is at $x = -2$. The graph has a removable discontinuity at $x = 2$.